



High pressure study of 1,3,5-trimethylbenzene oxidation

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ABSTRACT

The high pressure and high temperature kinetics of the 1,3,5-trimethylbenzene oxidation were investigated in the High Pressure Single Pulse Shock Tube at University of Illinois at Chicago. Experiments were performed at nominal reflected shock pressures of 20 and 50 atm, for three different equivalence ratios ($\phi = 0.51, 0.95$ and 1.86) and for a temperature range of 1017–1645 K. A variety of stable species ranging from aliphatic hydrocarbons to single ring and polycyclic aromatic hydrocarbons were sampled from the shock tube and analyzed using standard gas chromatographic techniques.

A detailed chemical kinetic model was developed to simulate the fuel and oxygen decay and the stable intermediate species profiles as obtained from the high pressure oxidation experiments. The model shows satisfactory predictions for the formation and consumption of most of the major intermediates.

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1. Introduction

Jet fuels constitute a major share of liquid transportation fuels and are used to power both the civilian and military aircrafts. The increase in air traffic over the recent years has led to an increased usage of jet fuels. Combustion of jet fuels produces emissions of carbon, nitrogen and sulfur species, volatile organic compounds and particulate matter. These emissions cause significant environmental damage and inhalation of particulate matter is hazardous to human health. The key element to reducing the emissions is in interpreting the combustion chemistry of aviation fuels. The coupling of detailed combustion kinetics and computational fluid dynamics will provide insight into effect of the fuel composition on pollutant formation and provide optimization parameters for increasing the efficiency of aviation gas turbine engines burning jet fuel.

Jet fuels consist of several hundreds of chemical components and significant compositional differences exist between different jet fuels such as Jet A-1, Jet-A and JP-8. Given the compositional differences and the multicomponent nature of jet fuels, building a detailed chemical kinetic model for wide spread predictive use with computational fluid mechanics is probably impossible. An alternative approach is to choose surrogate fuel components that replicate the parent fuel in both physical and chemical characteristics and consequently can provide a more general and thereby practical fuel model. This current work is a part of a collaborative effort that focuses on combustion of jet fuel surrogates and jet fuels

[1]. After considerable research, the collaborative effort has led to the selection of surrogate fuel components that represent different chemical classes of components of jet fuels. Among the alkylbenzene class of jet fuels, 1,3,5-trimethylbenzene (or mesitylene) has been chosen as the representative surrogate compound [2].

Prior experimental studies on 1,3,5-trimethylbenzene oxidation were limited to ignition delay times [3,4], diffusion flame extinction limits [5] and measurement of rate constants for hydrodealkylation of the side chains [6] and reaction with the hydroxyl radical [7,8]. Experimental data are also limited to low temperatures and pressures: temperatures of about 300–1000 K and pressures up to 21 atm. Detailed product distribution and mechanism of 1,3,5-trimethylbenzene oxidation are not available in literature. Hence the focus of this study is to provide the ‘speciation’ data for 1,3,5-trimethylbenzene oxidation for pressures between 20 and 55 atm and for temperatures up to 1650 K and to develop a kinetic model to simulate our experimental data. 1,3,5-Trimethylbenzene oxidation experiments were performed in the High Pressure Single Pulse Shock Tube (HPST) at UIC and a short description of the apparatus is given below.

2. Experimental and analytical section

The HPST has been described in detail in earlier publications [9,10]. The shock velocities were calculated from the response of pressure transducers (PCB model Nos. 113A23 [0–10000 psi] and 113B24 [0–1000 psi]) located in the side wall of the shock tube perpendicular to the incident shock wave. The reflected shock velocities were correlated to the temperature in the reaction zone by chemical thermometer technique [11]. Three chemical ther-

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monomers, cyclopropanecarbonitrile, 1,1,1-trifluoroethane and carbon disulfide were used. Cyclopropanecarbonitrile and 1,1,1-trifluoroethane were used for a combined temperature range of 901–1362 K. Carbon disulfide was used as thermometer for experiments in the temperature range of 1691–2033 K. For temperatures between 1362 K and 1691 K an interpolated temperature calibration was used. The uncertainty due to interpolated calibration fits for temperatures less than 1350 K and above 1350 K is $\pm 1\%$ and $\pm 2\%$ respectively. More details about the chemical thermometer technique are provided below.

The experimental parameters such as reaction time and pre-shock ($[th]_0$) and the post-shock ($[th]_f$) concentrations of the chemical thermometer are used to compute the thermometric T_5 values, using the Eq. (a).

$$T_5 = \frac{\left(-\frac{E_a}{R}\right)}{\ln} \left\{ \frac{\ln(1-x)}{At} \right\} \quad (a)$$

where x is the extent of reaction $x = \frac{([th]_0 - [th]_f)}{[th]_0}$, t is the reaction time, A and E_a are the pre-exponential factors and activation energies of decomposition reactions of the chemical thermometer. The reaction time is obtained from the pressure trace of the endwall pressure transducer. The reaction time is the time interval between arrival of the reflected shock wave at the end wall pressure transducer and the time at which reaction pressure reaches 80% of the maximum pressure due to quenching by the rarefaction wave. The pre-shock and post-shock concentrations of the chemical thermometers, cyclopropanecarbonitrile, 1,1,1-trifluoroethane and carbondisulfide were determined by the analytical apparatus. The analytical apparatus is described in detail in the last paragraph of this section.

The rate expressions for decomposition reactions of cyclopropanecarbonitrile, 1,1,1-Trifluoroethane and CS₂ are provided below

1. Cyclopropanecarbonitrile $\rightarrow$ c-CH₃CH=HCN (cis-crotonitrile), t-CH₃CH=HCN (trans-crotonitrile) and CH₂=CHCH₂CN (3-butenitrile), $k_1 = 3.148 \times 10^{15} \exp(-59208/RT)$ [12]. Eq. (a) was used to calculate the thermometric temperature T_5 .
2. 1,1,1-Trifluoroethane $\rightarrow$ CF₂=CH₂ (1,1-Difluoroethane)+HF, $k_2 = 4 \times 10^{15} \exp(-77493/RT)$ [13]. Eq. (a) was used to calculate the thermometric temperature T_5 .
3. CS₂ + M $\rightarrow$ CS + S + M (rxn1), CS₂ + S $\rightarrow$ CS + S₂ (rxn2), $k_{rxn1} = 2.51 \times 10^{14} \exp(-74000/RT)$ and $k_{rxn2} = 1 \times 10^{14} \exp(-4070/RT)$. The rate constants for these two reactions, rxn1 and rxn2 [14] were included in the chemical kinetic model and the CS₂ decomposition was simulated for a particular, temperature, pressure and the reaction time. The initial temperature was guessed and the simulations were performed until the temperature (T_s) at which the experimental CS₂ conversion matched the simulated CS₂ conversion. The temperature (T_s) was taken to be the thermometric temperature of the reflected shock.

The chemical thermometer results have been compared with the ideal shock wave equations [11]. The calculated values of T_5 values from ideal shock wave relations are higher when compared to the thermometric value of T_5 . The maximum deviation in the thermometric values is about 43 K in the ideal temperature range of 1100–1171 K, increases up to 88 K in the ideal temperature range of 1249–1423 K and increases up to 320 K for ideal temperatures up to 1960 K, for nominal reflected shock pressures of 50 atm.

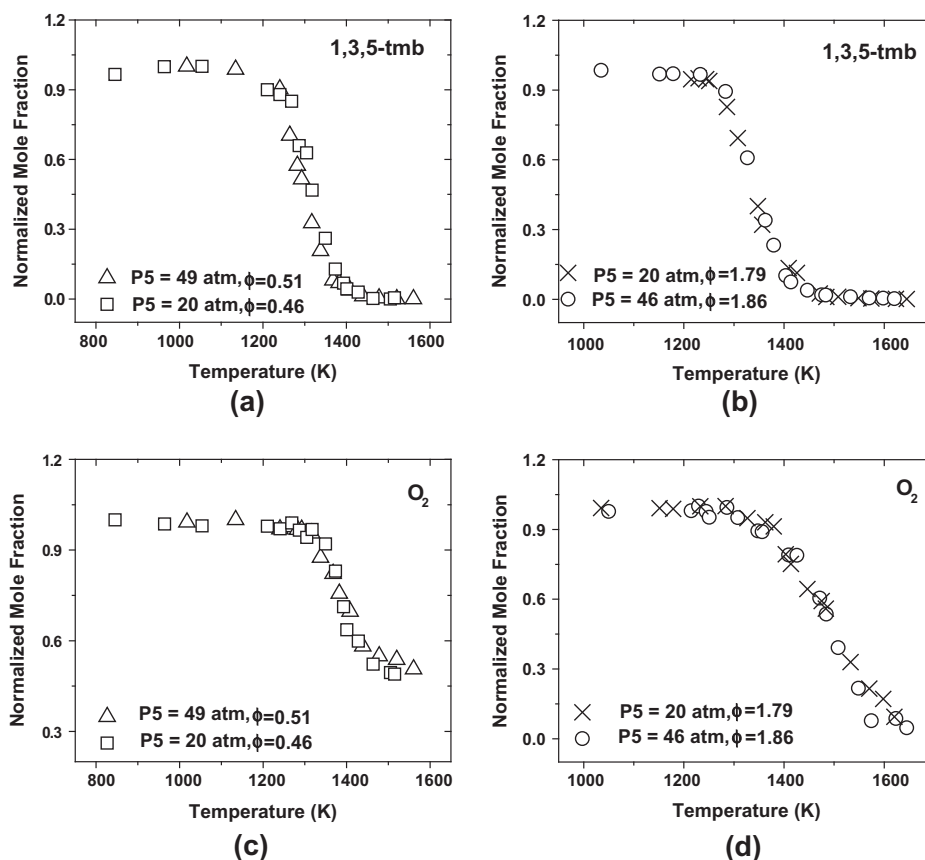


Fig. 1. Decay profiles of the fuel (1,3,5-tmb) and oxygen.

Reaction pressures were obtained from the pressure traces recorded from a high frequency pressure transducer mounted in the end wall of the driven section parallel to the long axis of the shock tube. The uncertainty in the measured reaction pressures and reaction times obtained from the recorded pressure profiles are 1% and 5% respectively.

Reagent mixtures consisting of 1,3,5-trimethylbenzene, ~80 ppm (99%, Sigma Aldrich) and krypton (99.999%, Specialty Gases of America) diluted in bath gas, argon (99.999%, Airgas) were prepared manometrically in heated tanks using a freeze and thaw procedure. Less than 200 ppm of the fuel, which was diluted in argon, was recommended to maintain isothermal conditions in the shock tube [9]. Low mole fractions of the fuel (<100 ppm) were used in the current experiments to maintain isothermal conditions. Such low concentrations minimize the temperature drop due to initial endothermic reactions and temperature rise due to later exothermic oxidation reactions, thereby maintaining isothermal conditions over the time range (1.5–2.5 ms) of the current experiments. Krypton was added as an internal standard in the experiments to account for any dilution of the post-shock gases by the driver gas, helium (99.998%, Airgas).

The shock tube and all the lines to the shock tube were heated to 100 °C. Over the course of the experiments, the fuel concentrations obtained at the temperatures ($T_5 < 1200$ K) where the fuel has not yet started to decay showed reproducible initial concentrations of the fuel. The reproducibility in the fuel concentrations at the lowest temperatures indicates that there were no fuel losses due to condensation on the walls of the shock tube or the gas sampling lines.

The gas samples were withdrawn from the center of the shock tube via an angled channel connected to the gas sampling port in

order to minimize the boundary layer effects. The boundary layer effects become prominent for low pressures around 6–10 atm, where the concentration of the fuel is observed to reach a low non-zero stable value even at very high temperatures. All the experiments discussed in this publication were performed for nominal reflected shock pressures above 15 atm where there are no boundary layer effects.

The gas samples from the shock tube were transferred to the analytical apparatus through a heated line (150 °C) and were analyzed by two GCs connected in series. The first GC (HP-6890) was used for the measurement of single ring and multi-ringed aromatic hydrocarbons and was equipped with a flame ionization detector (FID) connected to a DB-17MS column. The second GC (GC2) was used for the measurement of alkanes, alkenes, CO, CO₂, O₂ and inert gases. The separation and measurement of such gaseous species was attained with a HP-PLOT Q column connected to the FID. GC2 was also equipped with a Thermal Conductivity Detector (TCD) connected to a MOLSIEVE 5A column, which enabled the detection of O₂ and Krypton. Identification of reaction products was achieved by mass spectrometry (MS) with the aid of HP 5973 GC/MSD system. Uncertainties in species mole fractions are estimated to be no more than 5% for GC measurements of most of the major aliphatic and mono-aromatic hydrocarbons. The uncertainty is higher for multi-ring aromatic hydrocarbon species because the calibration curves for these species were deduced from the gas-phase calibration curve of benzene based on the relative ratio between the corresponding liquid-phase calibration curves. The liquid phase calibration curves were obtained using certified solutions of PAHs in appropriate solvents. As discussed in another publication from our laboratory, which dealt with the measurement of the polycyclic aromatic hydrocarbons from phenyl radical pyrolysis [12],

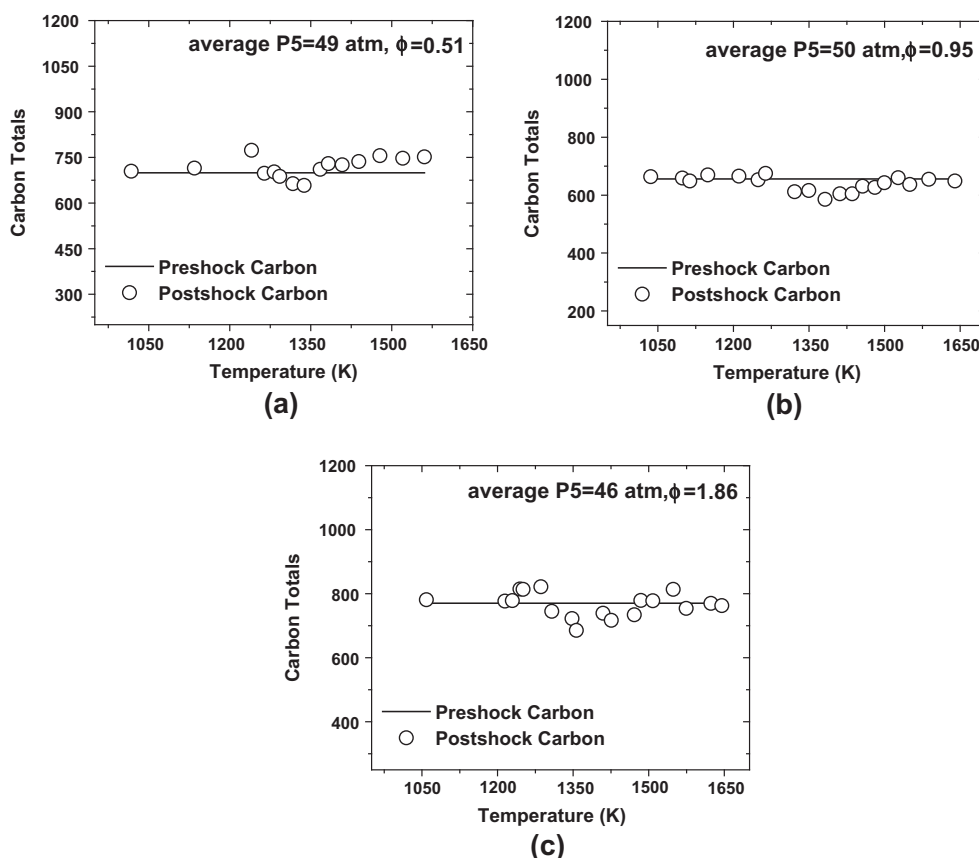


Fig. 2. Preshock and postshock carbon totals of 1,3,5-trimethylbenzene oxidation experimental datasets.

the maximum uncertainty in the measurement of C12 and C14 hydrocarbons is estimated to be 15–20% and 20–25% respectively. For even larger compounds ($>C_{14}$), the uncertainty could be as high as a factor of 2, since these calibrations were estimated from the calibration factors of three ringed species based on the differences in molecular weight. The calibration curves for diacetylene and triacetylene were deduced from experiments on acetylene decomposition (about 100 ppm in argon). Since the calibration for acetylene is well known from standard calibration mixtures, the calibration for diacetylene was obtained from the experiments where only acetylene and diacetylene were detected as major species. Similarly the calibration for triacetylene was deduced from experiments where acetylene and diacetylene were the major detected species [12].

3. Experimental results

1,3,5-Trimethylbenzene oxidation experiments were performed at fuel lean, stoichiometric and fuel rich conditions, over a temperature range of 1017–1645 K. Experiments at fuel lean and fuel rich conditions were performed at two nominal reflected shock pressures of 20 and 50 atm. All the experiments at stoichiometric conditions were conducted at an average pressure of 50 atm.

3.1. Effect of pressure on the fuel and oxidizer decay

The profiles of the fuel and oxidizer are shown as a function of temperature and concentration in Fig. 1a–d, for two different nominal reflected shock pressures of 20 and 50 atm and similar equivalence ratios ($\phi \approx 0.5$ and 2). The profiles of the fuel decay for the experimental datasets at fuel lean conditions and for different average pressures of 49 atm and 20 atm are shown in Fig. 1a. The data set at an average pressure of 49 atm and $\phi = 0.51$ is referred to as Set 1 and the dataset at an average pressure of 20 atm and $\phi = 0.46$ is referred to as Set 2. From Fig. 1a it can be observed that even though the equivalence ratios of the experimental datasets are slightly different the fuel decay profiles are similar. It is evident

from the figure that about 88% of the fuel remains unreacted at a temperature at 1241 K for Set 2, whereas Set 1 shows that 90% of the fuel remains unreacted at the same temperature. At slightly higher temperatures wherein almost 90% of the fuel has decayed, we can see that for Set 1 at a temperature of 1373 K, we see about 13% of the fuel is still unreacted. For a similar temperature of 1367 K for Set 2, we can observe that about 8% of the fuel is unreacted. The agreement in the concentration of the unreacted fuel remaining at similar temperatures for Set 1 and Set 2 is close to the error limits of the species concentration, which is $\pm 5\%$ for the fuel. Similar observation could be made for the fuel decay for the fuel rich oxidation experimental datasets and oxygen decay in the fuel lean oxidation experimental datasets, shown in Fig. 1b and c.

The fuel rich dataset at an average pressure of 46 atm and $\phi = 1.86$ is referred to as Set 3 and the fuel rich dataset at an average pressure of 20 atm and $\phi = 1.79$ is referred to as Set 4. The profiles of oxygen decay for Set 3 and Set 4 are shown in Fig. 1d. If we observe the oxygen decay profiles for Set 3 and Set 4, we can identify that at a temperature of 1362 K, 93% of oxygen is remains unreacted for Set 4, whereas at a similar temperature of 1356 K, about 89% of oxygen has still unreacted Set 3. At higher temperatures of 1620 K, about 9.4% of oxygen is remains unreacted for Set 4 and about 8.7% of oxygen remains unreacted for Set 3.

Even though there is a slight difference in equivalence ratios for the fuel lean and the fuel rich datasets, the decay profiles of the fuel and oxygen show similar values at similar temperatures, which are within the experimental uncertainties. Hence, we can conclude that over the pressure and temperature range of the current experiments, no significant pressure dependence was seen for the decay of the fuel and oxidizer.

3.2. Species measured and the experimental carbon balances

The species measured from all the experimental datasets ranged from species with carbon numbers from C1 to C16. The species measured were CO, CO₂, methane (CH₄), ethene (C₂H₄), ethane

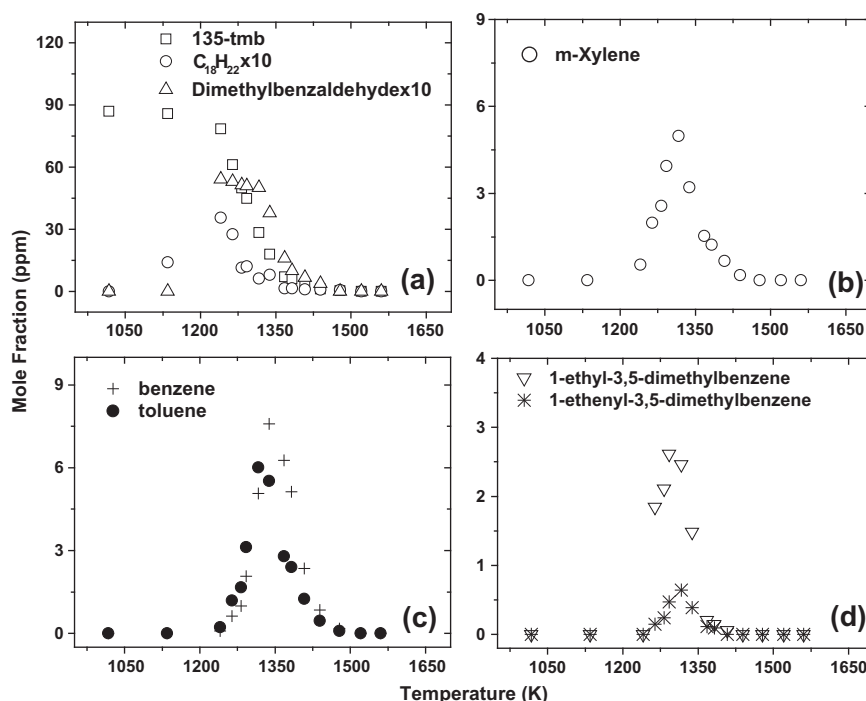


Fig. 3. Fuel decay and the formation of major intermediates, average $P_5 = 49.7$ atm, $\phi = 0.51$, reaction times = 1.26 to 2.8 ms.

(C₂H₆), acetylene (C₂H₂), propene (C₃H₆), allene (C₃H₄), propyne (C₃H₄), 1-butene (C₄H₈), 1,3-butadiene (C₄H₆–13), benzene (C₆H₆), toluene (C₆H₅CH₃), ethylbenzene (C₆H₅C₂H₅), m-xylene (C₈H₁₀), vinylacetylene (C₄H₄), diacetylene (C₄H₂), triacetylene (C₆H₂), o-xylene (C₈H₁₀), styrene (C₆H₅C₂H₃), phenylacetylene (C₆H₅C₂H), 1,2,4-trimethylbenzene (C₉H₁₂), 1-ethenyl-4-methylbenzene (C₉H₁₀), 1-ethenyl-3-methylbenzene (C₉H₁₀), 1-ethyl-3,5-dimethylbenzene (C₁₀H₁₄), indene (C₉H₈), 1-ethenyl-3,5-dimethylbenzene (C₁₀H₁₂), 3-methylbenzaldehyde (C₈H₈O), 1-methylindene (C₁₀H₁₀), 1-methyleneindene (C₁₀H₈), ethylbenzaldehyde (C₉H₁₀O), 3,5-dimethylbenzaldehyde (C₉H₁₀O), 1,2-dimethyl-4-(phenylmethyl)benzene (C₁₅H₁₆), 1,3,5-trimethylbiphenyl (C₁₅H₁₆), 1,2,4-trimethylbiphenyl (C₁₅H₁₆), 3,3'-dimethylbibenzyl (C₁₆H₁₈), 1,1'-(1,2-ethanediy)bis[3,5-dimethylbenzene] (C₁₈H₂₂), phenanthrene (C₁₄H₁₀), naphthalene (C₁₀H₈), acenaphthylene (C₁₂H₈), 1,1'-methylenebis-[3-methylbenzene] (C₁₅H₁₆), fluorene (C₁₃H₁₀), 2-phenylnaphthalene (C₁₆H₁₂), 9-methylanthracene (C₁₅H₁₂), 9-ethenylanthracene (C₁₆H₁₂), cyclopentaphenanthrene (C₁₅H₁₀), fluoranthene (C₁₆H₁₀), pyrene (C₁₆H₁₀), 2-methylfluoranthene (C₁₇H₁₂), 2,3-benzofluorene (C₁₇H₁₂), 1,8-dimethylnaphthalene (C₁₂H₁₂), 1,3-diethyl-5-methylbenzene (C₁₁H₁₆), 3,6-dimethylphenanthrene (C₁₆H₁₄), 4-methylphenanthrene (C₁₅H₁₂) and 3,3'-4,4'-tetramethylbiphenyl (C₁₆H₁₈).

The preshock and postshock carbon totals for the experimental data sets at fuel lean, stoichiometric and fuel rich oxidation datasets at a nominal pressure of 50 atm are shown in Fig. 2a–c. Almost 95% of carbon was recovered for the experimental dataset for $\phi = 0.51$. The maximum carbon losses for the experimental data sets at $\phi = 0.95$ and $\phi = 1.86$ are about 11% and 12% respectively. For the experimental datasets at stoichiometric and fuel rich conditions, the observed carbon loss is possibly due to the uncertainties in the calibration of polycyclic aromatic hydrocarbons which range from 15% to 25%.

From Fig. 2a–c, it can be observed that at a few temperatures, the postshock carbon totals are higher than the preshock carbon totals. The increased amounts of postshock carbon when compared to preshock carbon at temperatures lower than 1300 K, is due to the uncertainty in the measurement of 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene (C₁₈H₂₂). At temperatures higher than 1400 K, the uncertainties in the measurement of CO, CO₂, diacetylene and triacetylene species are responsible for the minor increase in postshock concentrations when compared the preshock concentrations.

3.3. Development of 1,3,5-trimethylbenzene oxidation mechanism based on the product distribution

A detailed mechanism or a chemical kinetic model for oxidation of 1,3,5-trimethylbenzene is not available in literature. However, mechanisms and chemical kinetic models are available for the oxidation of trimethylbenzenes. In these cases, the 1,2,3-, 1,2,4- and 1,3,5-trimethylbenzenes were lumped into a single species [15–17]. Since a detailed mechanism for the oxidation of 1,3,5-trimethylbenzene was not available in literature, the primary task was to develop a mechanism, a qualitative reaction pathway description, based on the product distribution observed from our experiments. The secondary task was to develop a chemical kinetic model, a predictive compilation of reactions with rate coefficients, based on the mechanism developed by us. In this section we explain the procedure followed in developing a mechanism for oxidation of the fuel based on the product distribution. In Section 4 we explain the rationale behind choosing the rate constants for the reactions included in the model based on the mechanism.

Since there was minor influence of the equivalence ratio on the species distribution, the oxidation experiments at $\phi = 0.5$, have been chosen to define the reaction mechanism of the fuel based

on the species observed at different temperatures. The concentration profiles of the fuel and the major intermediates are provided in Fig. 3a–d. At lower temperatures ($T_5 = 1134$ K), for the decay of 1,3,5-trimethylbenzene we observed the formation of 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene (C₁₈H₂₂). At a temperature of about 1181 K we observed the formation of 3,5-dimethylbenzaldehyde (see Fig. 3a). At slightly higher temperatures ($T_5 = 1264$ K) we observed the formation of m-xylene, methane, ethane, 1-ethyl-3,5-dimethylbenzene, 1-ethenyl-3,5-dimethylbenzene, toluene and benzene (see Fig. 3b–d).

Based on the product distribution observed from our experiments we can safely assume that 1,3,5-trimethylbenzene could be oxidized by sequential oxidation and removal of the methyl side chains, similar to the mechanism of m-xylene oxidation as proposed by Emdee et al. [18].

The procedure followed by us, in developing the chemical kinetic model for the oxidation of 1,3,5-trimethylbenzene is mentioned in the following section (Section 4). In Section 5, the modeling results are compared against the experimental data and reaction path analysis is performed to get a better understanding of the decay of the fuel and formation of the intermediates and that also serves as a means to validate our proposed mechanism.

4. UIC 1,3,5-trimethylbenzene oxidation model

4.1. Kinetics for 1,3,5-trimethylbenzene decay and formation of major single ringed aromatic intermediates

The UIC 1,3,5-trimethylbenzene oxidation model is developed by adding reactions of methyl side chains oxidation and abstraction to our UIC m-Xylene Oxidation Model [19] (see Table 1).

Rate constants for the oxidation reactions of 1,3,5-trimethylbenzene were directly taken from literature, if they were available. The list of the rate constants which were taken directly from literature is provided in Table 2. The rate constant for C–H bond split of 1,3,5-trimethylbenzene from the methyl side chains was determined by Brand et al. [20] through laser excitation experiments (R1, R-1 in Table 2). The rate constants for the abstraction of a hydrogen atom from the methyl side chain and displacement of a methyl side chain by a hydrogen atom were adapted from the experimental study of Tsang et al. [21] (R2 and R3 in Table 2).

Table 1
1,3,5-Trimethylbenzene oxidation experimental conditions.

Average shock pressure (atm)	Fuel (ppm)	Temperature range (K)	Φ	Reaction time (ms)
20.2	77	1033–1515	0.46	1.55–2.55
49.7	86	1017–1560	0.51	1.26–2.80
50.2	73	1035–1639	0.95	1.13–1.69
19.7	81	1151–1620	1.79	1.40–2.47
45.7	87	1058–1645	1.86	1.11–2.11

Table 2
Rate constants of reactions taken directly from literature.^a

Rxn	Reaction ^a	A	n	E	Ref.
R1	C ₉ H ₁₂ $\Rightarrow$ C ₉ H ₁₁ + H	2.24E+16	0	87237	[20]
R-1	C ₉ H ₁₁ + H $\Rightarrow$ C ₉ H ₁₂	7.80E+13	0	0	[20]
R2	C ₉ H ₁₂ + H $\Rightarrow$ C ₉ H ₁₁ + H ₂	3.70E+14	0	8628	[21]
R3	C ₉ H ₁₂ + H $\Rightarrow$ MXYLENE+CH ₃	6.70E+13	0	6468	[21]
R4	C ₉ H ₁₂ + OH $\Rightarrow$ C ₉ H ₁₁ + H ₂ O	2.39E+12	0	–1700	[23]
R5	C ₉ H ₁₁ + C ₉ H ₁₁ $\Rightarrow$ C ₁₈ H ₂₂	7.94E+12	0	0	[24]
R-5	C ₁₈ H ₂₂ $\Rightarrow$ C ₉ H ₁₁ + C ₉ H ₁₁	1.00E+15	0	62988	[24]

^a Rate constant, $k = AT^n \exp(-Ea/RT)$; Rate constant (units: mol, s, cm³, cal).

Table 3

Rate constants for the decay of the fuel and the formation and consumption of other major intermediates.

Rxn	Reaction ^a	A	n	Ea	Ref.
R6	$C_9H_{12} = C_6H_3(CH_3)_2 + CH_3$	$3.37E+17$	0	100437	[55] ^b
R-6	Reverse	$1.38E+13$	0	45	[20] ^c
R-2	$C_9H_{11} + H_2 = C_9H_{12} + H$	$2.82E+12$	0	14500	[26] ^d
R7	$C_9H_{12} + CH_3 = C_9H_{11} + CH_4$	$3.51E+11$	0	7700	[27] ^e
R-7	Reverse	$1.20E+13$	0	27100	[27] ^f
R-4	$C_9H_{11} + H_2O = C_9H_{12} + OH$	$1.23E+10$	0.7003	29538	[28] ^h
R8	$C_9H_{12} + O_2 = C_9H_{11} + HO_2$	$3.42E+07$	2.5	44946	[28] ^j
R-8	Reverse	$1.44E+07$	1.974	4085	[28] ^j
R-3	$MXYLENE+CH_3 = C_9H_{12} + H$	$8.00E+11$	0	15900	[29] ^k
R9	$C_9H_{11} + CH_3 = DIMETB$	$1.10E+12$	0	0	[20] ^l
R-9	Reverse	$8.00E+14$	0	80922	[20] ^b
R10	$DIMETB + H = DIMCHCH_3 + H_2$	$7.26E+07$	2	5345	[28] ^m
R-10	Reverse	$1.98E+06$	2.2	22404	[28] ⁿ
R11	$DIMETB + H = DIMCH_2CH_2 + H_2$	$1.59E+08$	1.5	7412	[28] ^p
R-11	Reverse	$6.51E+05$	1.564	8720	[28] ^p
R12	$DIMETB + O = DIMCHCH_3 + OH$	$4.91E+13$	0	3795	[28] ^q
R-12	Reverse	$3.19E+11$	0.18	19027	[28] ^r
R13	$DIMETB + HO_2 = DIMCHCH_3 + H_2O_2$	$1.36E+05$	2.5	13522	[28] ^s
R-13	Reverse	$9.75E+05$	2.04	14045	[28] ^t
R14	$DIMETB + HO_2 = DIMCH_2CH_2 + H_2O_2$	$7.81E+04$	2.5	16849	[28] ^u
R-14	Reverse	$2.98E+03$	1.901	1621	[28] ^v
R15	$DIMETB + OH = DIMCH_2CH_2 + H_2O$	$4.80E+12$	0	0	[28] ^w
R-15	Reverse	$8.51E+10$	-0.03	16598	[28] ^x
R16	$C_6H_3(CH_3)_2 + C_2H_4 = DIMCH_2CH_2$	$4.04E+03$	2.64	1459	[30] ^y
R-16	Reverse	$1.72E+11$	0.78	38704	[30] ^z
R17	$DIMCH_2CH_2 = DIMSTYR + H$	$1.83E+06$	2.081	33207	[30] ^{aa}
R-17	Reverse	$1.63E+06$	2.158	1994	[30] ^{ab}
R18	$DIMCH_2CH_2 = DIMCHCH_3$	$5.96E+05$	2.074	29582	[30] ^{ac}
R-18	Reverse	$2.30E+07$	1.737	44456	[30] ^{ad}
R19	$DIMCHCH_3 = DIMSTYR + H$	$3.74E+08$	1.55	44609	[30] ^{ae}
R-19	Reverse	$1.19E+07$	1.925	-108	[30] ^{af}
R20	$DIMSTYR + H = DIMCHCH + H_2$	$1.99E+06$	2.53	12239	[31] ^{ag}
R-20	Reverse	$1.98E+06$	2.2	22404	[28] ^{ah}
R21	$DIMSTYR + O = C_6H_3(CH_3)_2 + CH_2HCO$	$3.50E+13$	0	2832	[31] ^{ai}
R22	$C_6H_3(CH_3)_2 + C_2H_2 = DIMCHCH$	$2.69E+06$	2.05	3720	[56] ^{aj}
R-22	Reverse	$1.35E+14$	0.34	45710	[56] ^{ak}
R23	$DIMETB + H = MCPHC_2H_5 + CH_3$	$5.80E+13$	0	8100	[15]
R-23	Reverse	$1.20E+12$	0	15900	[15]
R24	$DIMSTYR + H = MCSTYREN + CH_3$	$5.80E+13$	0	8100	[15] ^{al}
R-24	Reverse	$1.20E+12$	0	15900	[15] ^{am}
R25	$DIMETB + H = MXYLENE + C_2H_5$	$5.20E+13$	0	4100	[15]
R-25	Reverse	$1.20E+11$	0	15900	[15]
R26	$DIMSTYR + H = MXYLENE + C_2H_3$	$5.20E+13$	0	4100	[15] ^{an}
R-26	Reverse	$1.20E+11$	0	15900	[15] ^{ao}
R27	$C_9H_{11} + O_2 = C_9H_{11}OO$	$3.08E+11$	-0.01	751.8	[57] ^{ap}
R-27	Reverse	$7.85E+13$	0	10224	[57] ^{aq}
R28	$C_9H_{11}OO \Rightarrow DIMPHHCO + OH$	$1.80E+12$	0	10066	[57] ^{ar}
R29	$C_9H_{11}OO = OC_6H_3(CH_3)_2 + CH_2O$	$5.55E+09$	0	17341	[57] ^{as}
R-29	Reverse	$1.00E+13$	0	0	[33] ^b
R30	$DIMPHHCO = DIMPHCO + H$	$3.98E+15$	0	83661	[34] ^{at}
R30	Reverse	$3.00E+13$	0	0	[33] ^b
R31	$DIMPHCO = C_6H_3(CH_3)_2 + CO$	$1.25E+15$	0	27646	[35] ^{au}
R-31	Reverse	$1.17E+12$	0	2778	[58] ^{av}
R32	$DIMPHHCO + O_2 = DIMPHCO + HO_2$	$7.14E+13$	0	38950	[59] ^{aw}
R-32	Reverse	$3.01E+13$	0	0	[36] ^{ax}
R33	$DIMPHHCO + OH = DIMPHCO + H_2O$	$1.20E+10$	1.18	-447	[59] ^{ay}
R-33	Reverse	$5.14E+11$	1.35	26000	[36] ^{az}
R34	$DIMPHHCO + H = DIMPHCO + H_2$	$3.50E+14$	0	4928	[59] ^{ba}
R-34	Reverse	$1.31E+11$	1.82	17600	[36] ^{bb}
R35	$DIMPHHCO + O = DIMPHCO + OH$	$6.33E+13$	0	3080	[59] ^{bc}
R-35	Reverse	$4.22E+13$	0	0	[36] ^{bd}
R36	$DIMPHHCO + H = MCPHCHCO + CH_3$	$5.80E+13$	0	8100	[15] ^{al}
R-36	Reverse	$1.20E+12$	0	15900	[15] ^{am}
R37	$C_9H_{11} + O = DIMPHHCO + H$	$4.00E+13$	0	0	[37] ^{be}
R-37	Reverse	$3.98E+13$	0	102540	[38] ^{bf}
R38	$C_9H_{11} + OH \Rightarrow DIMPHCH_2OH$	$2.00E+13$	0	0	[37] ^{bg}
R38	$DIMPHCH_2OH + OH \Rightarrow DIMPHCH_2O + H_2O$	$5.00E+12$	0	0	[37] ^{bh}
R39	$DIMPHCH_2OH + H = MCPCH_2OH + CH_3$	$5.80E+13$	0	8100	[15] ^{al}
R-39	Reverse	$1.20E+12$	0	15900	[15] ^{am}
R40	$C_9H_{11} + HO_2 \Rightarrow C_9H_{11}OOH$	$8.00E+13$	0	0	[39] ^{bi}
R41	$C_9H_{11}OOH \Rightarrow DIMPHCH_2O + OH$	$3.29E+13$	0.42	39890	[39] ^{bj}
R42	$C_9H_{11} + HO_2 \Rightarrow DIMPHCH_2O + OH$	$3.86E+10$	0	1456	[40] ^{bk}
R43	$DIMPHCH_2O \Rightarrow DIMPHHCO + H$	$5.07E+08$	1.56	16850	[41] ^{bl}
R44	$DIMPHCH_2O \Rightarrow C_6H_3(CH_3)_2 + CH_2O$	$1.09E+14$	0.157	31160	[41] ^{bm}

(continued on next page)

Table 3 (continued)

Rxn	Reaction ^a	A	n	Ea	Ref.
R45	DIMPHCH ₂ O ⇒ MXYLENE + HCO	1.81E+13	0	22 717	[41] ^{bn}
R46	C ₆ H ₃ (CH ₃) ₂ + O ₂ = OC ₆ H ₃ (CH ₃) ₂ + O	8.57E+18	−2.27	7189	[60] ^{bo}
R-46	Reverse	2.21E+25	−3.39	16945	[60] ^{bp}
R47	MCPHC ₂ H ₅ + H = ΦC ₂ H ₅ + CH ₃	1.70E+14	0	8100	[15]
R-47	Reverse	1.20E+12	0	15900	[15]
R48	MCSTYREN + H = ΦCH ₃ + C ₂ H ₃	5.20E+13	0	4100	[15] ^{bq}
R-48	Reverse	1.20E+11	0	15900	[15] ^{br}
R49	MCSTYREN + H = ΦC ₂ H ₃ + CH ₃	1.70E+14	0	8100	[15] ^{bs}
R-49	Reverse	1.20E+12	0	15900	[15] ^{bt}
R50	MCSTYREN = C ₆ CH ₄ CH ₃ + C ₂ H ₃	8.00E+15	0	114 177	[55] ^b
R-50	Reverse	2.00E+13	0	0	[33] ^b
R51	MCPHC ₂ H ₅ = C ₆ CH ₄ CH ₃ + C ₂ H ₅	8.00E+15	0	100 907	[55] ^b
R-51	Reverse	2.00E+13	0	0	[33] ^b
R52	DIMPHHCO + H = MXYLENE + HCO	5.20E+13	0	4100	[15] ^{bq}
R-52	Reverse	1.20E+11	0	15900	[15] ^{br}
R53	DIMPHHCO = C ₆ H ₃ (CH ₃) ₂ + HCO	8.00E+15	0	147 206	[55] ^b
R-53	Reverse	2.00E+13	0	0	[33] ^b
R54	DIMPHCH ₂ OH = C ₆ H ₃ (CH ₃) ₂ + CH ₂ OH	8.00E+15	0	148 412	[55] ^b
R-54	Reverse	2.00E+13	0	0	[33] ^b
R55	C ₆ H ₃ (CH ₃) ₂ + H ⇒ MXYLENE	1.00E+14	0	0	[33] ^b
R56	C ₆ H ₃ (CH ₃) ₂ + O ⇒ OC ₆ H ₃ (CH ₃) ₂	1.00E+14	0	0	[33] ^b
R57	C ₆ H ₃ (CH ₃) ₂ + OH ⇒ HOC ₆ H ₃ (CH ₃) ₂	1.00E+13	0	0	[33] ^b

^a Rate constant, $k = AT^n \exp(-Ea/RT)$; rate constant (units: mol, s, cm³, cal), Φ denotes the phenyl ring.

^b Estimated using rate rules proposed by Dean [55] and Allara and Shaw [33].

^c k same as Φ + CH₃ = ΦCH₃.

^d k same as ΦCH₂ + H₂ = ΦCH₃ + H.

^e A-factor × 3 of CH₃ + ΦCH₃ = CH₄ + ΦCH₂.

^f k same as ΦCH₂ + CH₄ = ΦCH₃ + CH₃.

^g A-factor × 3 of OH + ΦCH₃ = H₂O + ΦCH₂.

^h k same as ΦCH₂ + H₂O = ΦCH₃ + OH.

ⁱ A-factor × 3 of O₂ + ΦCH₃ = HO₂ + ΦCH₂.

^j k same as ΦCH₂ + HO₂ = ΦCH₃ + O₂.

^k A-factor × (2/3) of CH₃ + C₆H₆ = ΦCH₃ + H.

^l A-factor × (1/10) of ΦCH₂ + CH₃ = ΦC₂H₅, adjusted based on the agreement of the modeling results with the experimental data.

^m A-factor × 2.2 of ΦC₂H₅ + H = ΦCHCH₃ + H₂.

ⁿ k same as ΦCHCH₃ + H₂ = ΦC₂H₅ + H.

^o A-factor × 2.2 of ΦC₂H₅ + H = ΦCH₂CH₂ + H₂.

^p k same as ΦCH₂CH₂ + H₂ = ΦC₂H₅ + H.

^q A-factor × 2.2 of ΦC₂H₅ + O = ΦCHCH₃ + OH.

^r k same as ΦCHCH₃ + OH = ΦC₂H₅ + O.

^s A-factor × 2.2 of ΦC₂H₅ + HO₂ = ΦCHCH₃ + H₂O₂.

^t k same as ΦCHCH₃ + HO₂ = ΦC₂H₅ + HO₂.

^u A-factor × 2.2 of ΦC₂H₅ + HO₂ = ΦCH₂CH₂ + H₂O₂.

^v k same as ΦC₂H₅ + HO₂ = ΦCHCH₃ + HO₂.

^w k same as ΦC₂H₅ + OH = products.

^x k same as ΦCH₂CH₂ + HO₂ = ΦC₂H₅ + OH.

^y k same as Φ + C₂H₄ = ΦCH₂CH₂.

^z k same as ΦCH₂CH₂ = Φ + C₂H₄.

^{aa} k same as ΦCH₂CH₂ = ΦC₂H₃ + H.

^{ab} k same as ΦC₂H₃ + H = ΦCH₂CH₂.

^{ac} k same as ΦCH₂CH₂ = ΦCHCH₃.

^{ad} k same as ΦCHCH₃ = ΦCH₂CH₂.

^{ae} k same as ΦCHCH₃ = ΦC₂H₃ + H.

^{af} k same as ΦC₂H₃ + H = ΦCHCH₃.

^{ag} A-factor × 3 of ΦC₂H₃ + H = ΦCHCH + H₂.

^{ah} k same as ΦCH₂CH₂ + H₂ = ΦC₂H₅ + H.

^{ai} k same as ΦC₂H₃ + O = Φ + CH₂HCO.

^{aj} k same as Φ + C₂H₂ = ΦC₂H₃.

^{ak} k same as ΦC₂H₃ = Φ + C₂H₂.

^{al} k same as DIMETB + H = MCPHC₂H₅ + CH₃.

^{am} k same as MCPHC₂H₅ + CH₃ = DIMETB + H.

^{an} k same as DIMETB + H = MXYLENE + C₂H₅.

^{ao} k same as MXYLENE + C₂H₅ = DIMETB + H.

^{ap} k same as C₈H₉ + O₂ = C₈H₉OO.

^{aq} k same as C₈H₉OO = C₈H₉ + O₂.

^{ar} A-factor × (1/10) of C₈H₉OO = CH₃C₆H₄CHO + OH, adjusted based on the agreement of the modeling results with the experimental data.

^{as} k same as OC₆H₃(CH₃)₂ + CH₂O = C₉H₁₁OO.

^{at} k same as ΦCHO = ΦCO + H.

^{au} k same as ΦCO = Φ + CO.

^{av} k same as Φ + CO = ΦCO.

^{aw} A-factor × 7 of ΦCHO + O₂ = ΦCO + HO₂.

^{ax} k same as HCO + HO₂ = products.

^{ay} A-factor × 7 of ΦCHO + OH = ΦCO + H₂O.

^{az} k same as HCO + HO₂ = CH₂O + OH.

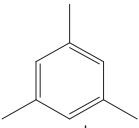
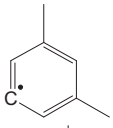
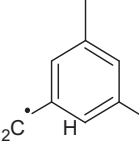
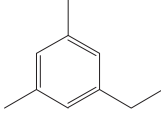
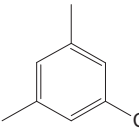
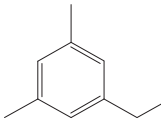
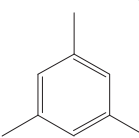
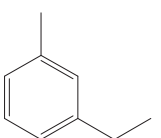
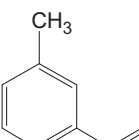
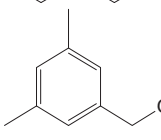
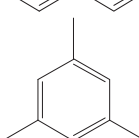
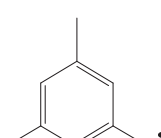
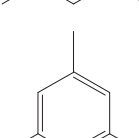
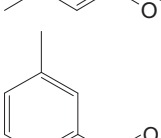
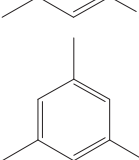
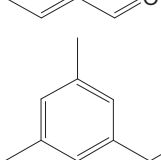
^{ba} A-factor × 7 of ΦCHO + H = ΦCO + H₂.

^{bb} k same as CH₃CO + H₂ = CH₃CHO + H.

- ^{bc} A-factor $\times 7$ of $\Phi\text{CHO} + \text{O} = \Phi\text{CO} + \text{OH}$.
^{bd} k same as $\text{CH}_3\text{CO} + \text{OH} = \text{products}$.
^{be} k same as $\Phi\text{CH}_2 + \text{O} = \Phi\text{CHO} + \text{H}$.
^{bf} k same as $\text{HCO} + \text{H} = \text{O} + \text{CH}_2$.
^{bg} k same as $\Phi\text{CH}_2 + \text{OH} = \Phi\text{CH}_2\text{OH}$.
^{bh} k same as $\Phi\text{CH}_2\text{OH} + \text{OH} = \Phi\text{CH}_2\text{O} + \text{H}_2\text{O}$.
^{bi} k same as $\Phi\text{CH}_2 + \text{HO}_2 = \Phi\text{CH}_2\text{OOH}$.
^{bj} k same as $\Phi\text{CH}_2\text{OOH} = \Phi\text{CH}_2\text{O} + \text{OH}$.
^{bk} k same as $\Phi\text{CH}_2 + \text{HO}_2 = \Phi\text{CH}_2\text{O} + \text{OH}$.
^{bl} k same as $\Phi\text{CH}_2\text{O} = \Phi\text{CHO} + \text{H}$.
^{bm} k same as $\Phi\text{CH}_2\text{O} = \Phi + \text{CH}_2\text{O}$.
^{bn} k same as $\Phi\text{CH}_2\text{O} = \text{C}_6\text{H}_6 + \text{CHO}$.
^{bo} A-factor $\times (1/100)$ of $\text{C}_6\text{H}_4\text{CH}_3 + \text{O}_2 = \text{OC}_6\text{H}_4\text{CH}_3 + \text{O}$, adjusted based on the agreement of the modeling results with the experimental data.
^{bp} A-factor $\times (1/100)$ of $\text{OC}_6\text{H}_4\text{CH}_3 + \text{O} = \text{C}_6\text{H}_4\text{CH}_3 + \text{O}_2$, adjusted based on the agreement of the modeling results with the experimental data.
^{bq} k same as $\text{MCPHCH}_5 + \text{H} = \Phi\text{CH}_3 + \text{C}_2\text{H}_5$.
^{br} k same as $\Phi\text{CH}_3 + \text{C}_2\text{H}_5 = \text{MCPHCH}_5 + \text{H}$.
^{bs} k same as $\text{MCPHC}_2\text{H}_5 + \text{H} = \Phi\text{C}_2\text{H}_5 + \text{CH}_3$.
^{bt} k same as $\Phi\text{C}_2\text{H}_5 + \text{CH}_3 = \text{MCPHC}_2\text{H}_5 + \text{H}$.

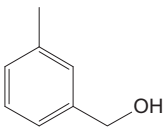
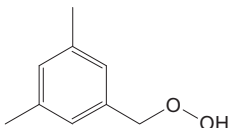
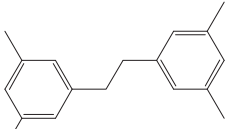
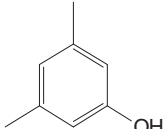
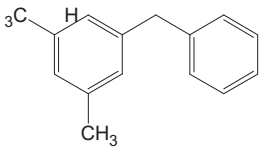
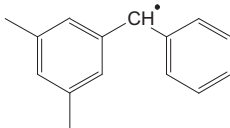
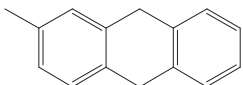
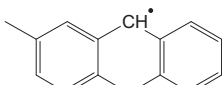
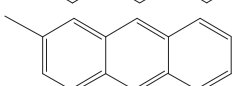
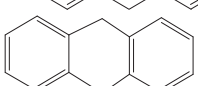
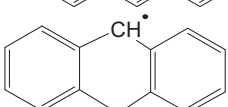
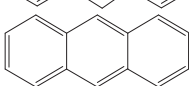
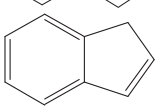
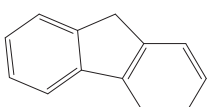
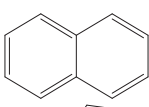
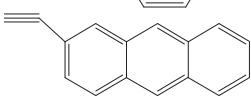
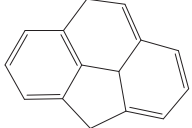
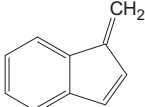
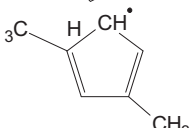
Table 4

Structures of species present in reactions in Tables 3 and 5, discussed in the text.

Species name in model	Species structure	Species name in model	Species structure
C_9H_{12}		$\text{C}_6\text{H}_3(\text{CH}_3)_2$	
C_9H_{11}		DIMETB	
DIMCHCH ₃		DIMCH ₂ CH ₂	
DIMSTYR		MCPHC ₂ H ₅	
MCSTYREN		$\text{C}_9\text{H}_{11}\text{OO}$	
DIMPHHCO		$\text{OC}_6\text{H}_3(\text{CH}_3)_2$	
DIMPHCO		MCPHHCO	
DIMPHCH ₂ OH		DIMPHCH ₂ O	

(continued on next page)

Table 4 (continued)

Species name in model	Species structure	Species name in model	Species structure
MCPCH ₂ OH		C ₉ H ₁₁ OOH	
C ₁₈ H ₂₂		HOC ₆ H ₃ (CH ₃) ₂	
DIMERS196		RDIMERS196a	
PTRIARO192		PTRIARO192a	
A ₃ CH ₃		PTRIARO178	
RPTRIARO178		A ₃	
INDENE		FLUORENE	
A ₂		A ₃ C ₂ H	
Cyclopentaphenanthrene		Methyleneindene	
Dimethylcyclopentadienyl radical (DIMCPD)			

The OH initiated oxidation of 1,3,5-trimethylbenzene was studied experimentally and the products of this reaction were quantified by Smith et al. [22] in a fixed volume dynamic reactor operated at atmospheric pressure and a temperature of 298 K. The primary products of OH + 1,3,5-trimethylbenzene as measured by Smith et al. were methylglyoxal, 3-methyl-5-methylidene-5(2H)-2-furanone, 3,5-dimethyl-3(2H)-2-furanone, 3,5-dimethyl-5(2H)-2-furanonebiacetyl and 2-methyl-4-oxo-2-pentenal. We have not observed the formation of any of these products from our experiments, the reason for this being that the experimental study of Smith et al. was conducted at much lower pressure and temperatures when compared to our experimental conditions. In a much earlier study by Perry et al. [23], the authors measured the absolute rate constants for the reaction of OH radicals with 1,3,5-trimethylbenzene over the temperature range of 296–473 K using a flash photolysis-resonance technique. For the reaction of OH with 1,3,5-trimethylbenzene the initial reaction can be either abstraction of the methyl side chain hydrogen atoms forming 1,3,5-trimethylbenzyl radical or addition of OH to the aromatic ring. The authors have mentioned that at temperatures above 380 K, the abstraction reaction is the sole reaction observed due to the rapid decomposition of the OH-aromatic adduct. Hence the steps for formation of OH-aromatic adduct is not included in our model and only the abstraction step is included. The rate constant of the abstraction reaction is taken from Perry et al. [23] (R4 in Table 2). The rate constant for the formation of 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene (C₁₈H₂₂) from recombination of 1,3,5-trimethylbenzyl radicals was taken from 1,7-octadiene pyrolysis study of Tsang and Walker. The authors used 1,3,5-trimethylbenzene as an inhibitor which captured more reactive fragments formed in the decomposition of 1,7-octadiene (R5 and R-5 in Table 2).

Other than the reactions mentioned in Table 2, the rate constants for all the other reactions in the oxidation of 1,3,5-trimethylbenzene were estimated from analogous oxidation reactions of toluene, and were adjusted for degeneracy, because direct experi-

mentary data were not available. The authors have mentioned that at temperatures above 380 K, the abstraction reaction is the sole reaction observed due to the rapid decomposition of the OH-aromatic adduct. Hence the steps for formation of OH-aromatic adduct is not included in our model and only the abstraction step is included. The rate constant of the abstraction reaction is taken from Perry et al. [23] (R4 in Table 2). The rate constant for the formation of 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene (C₁₈H₂₂) from recombination of 1,3,5-trimethylbenzyl radicals was taken from 1,7-octadiene pyrolysis study of Tsang and Walker. The authors used 1,3,5-trimethylbenzene as an inhibitor which captured more reactive fragments formed in the decomposition of 1,7-octadiene (R5 and R-5 in Table 2).

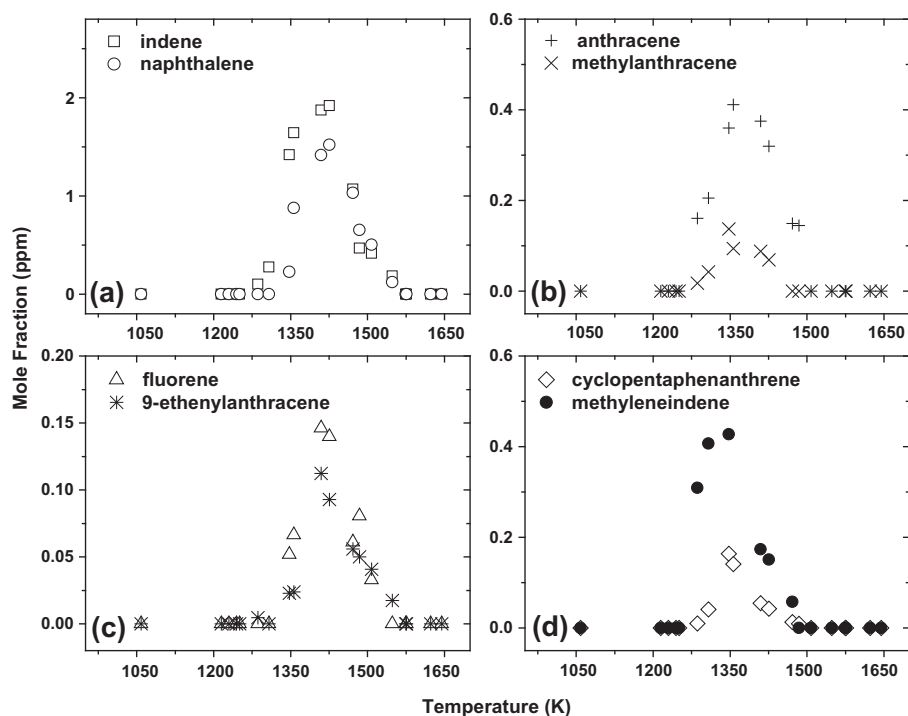


Fig. 4. Major polycyclic aromatic hydrocarbons formed in 1,3,5-trimethylbenzene oxidation experiments, average $P_5 = 45.6$ atm, $\phi = 1.86$ and reaction time = 1.11 to 2.11 ms.

Table 5

Reactions for the formation of $C_{18}H_{22}$, DIMERS196, anthracene and methyl anthracene.

Rxn	Reaction ^a	A	n	Ea	Ref.
R58	$C_9H_{11} + C_6H_6 = \text{DIMERS196} + H$	$1.99E+11$	0	3994	[62] ^b
R-59	Reverse	$1.99E+14$	0	29400	[63]
R59	$\text{DIMERS196} + C_9H_{11} = \text{RDIMERS196a} + C_9H_{12}$	$1.60E+11$	0	12800	[15]
R-59	Reverse	$1.60E+09$	0	12800	[15]
R60	$\text{DIMERS196} + H = \text{RDIMERS196a} + H_2$	$4.00E+15$	0	8400	[15]
R-60	Reverse	$2.80E+11$	0	14500	[15]
R61	$\text{DIMERS196} + CH_3 = \text{RDIMERS196a} + CCH_4$	$1.60E+12$	0	8800	[15]
R-61	Reverse	$6.60E+11$	0	27500	[15]
R62	$\text{RDIMERS196a} = \text{PTRIARO192} + H$	$4.20E+11$	0	23500	[15]
R-62	Reverse	$5.80E+13$	0	29400	[15]
R63	$\text{RDIMERS196a} = \text{PTRIARO178} + CH_3$	$8.40E+11$	0	23500	[15]
R-63	Reverse	$1.20E+12$	0	29400	[15]
R64	$\text{PTRIARO192} + H = \text{RPTRIARO192a} + H_2$	$4.00E+15$	0	8400	[15]
R-64	Reverse	$2.80E+11$	0	14500	[15]
R65	$\text{PTRIARO178} + H = \text{RPTRIARO178a} + H_2$	$4.00E+15$	0	8400	[15]
R-65	Reverse	$2.80E+11$	0	14500	[15]
R66	$\text{PTRIARO192} + CH_3 = \text{RPTRIARO192a} + CH_4$	$1.60E+12$	0	8800	[15]
R-66	Reverse	$6.60E+11$	0	27500	[15]
R67	$\text{PTRIARO178} + CH_3 = \text{RPTRIARO178a} + CH_4$	$1.60E+12$	0	8800	[15]
R-67	Reverse	$6.60E+11$	0	27500	[15]
R68	$\text{RPTRIARO178a} = A_3 + H$	$3.50E+13$	0	28400	[15]
R-68	Reverse	$4.30E+13$	0	4500	[15]
R69	$\text{RPTRIARO192a} = A_3CH_3 + H$	$3.50E+13$	0	28400	[15]
R-69	Reverse	$4.30E+13$	0	4500	[15]
R70	$\text{DIMERS196} + H = \text{DIMERS182} + CH_3$	$6.70E+13$	0	6468	[21] ^c
R-70	Reverse	$1.20E+12$	0	15900	[15]
R71	$\text{DIMERS196} + H = \text{DIMERS182} + CH_3$	$6.70E+13$	0	6468	[21] ^c
R72	Reverse	$1.20E+12$	0	15900	[15]
R73	$\text{DIMERS182} + H = C_6H_5CH_2C_6H_5 + CH_3$	$6.70E+13$	0	6468	[21] ^c
R-73	Reverse	$1.20E+12$	0	15900	[15]

^a $k = AT^n \exp(-E_a/RT)$: rate constant (units: mol, s, cm³, cal).

^b A-factor same as $C_9H_{12} + H = \text{MXYLENE} + CH_3$.

^c A-factor $\times (1/2) C_{12}H_{10} + H = \Phi + C_6H_6$.

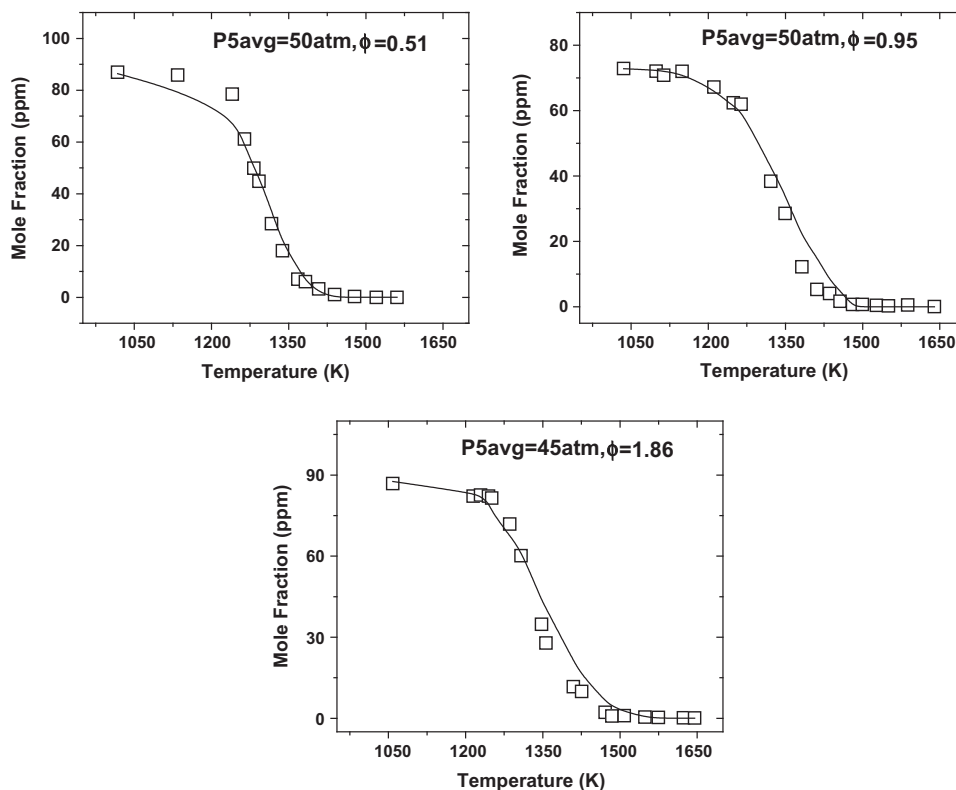
mental measurements of these rate constants were not available in literature. For example, the rate constant for abstraction of the methyl side chain hydrogen atoms of 1,3,5-trimethylbenzene by a methyl radical was estimated from the analogous reaction of a

methyl radical abstracting a hydrogen atom from toluene. The pre-exponential factor was multiplied by a factor of 3, in order to account for the number of abstractable hydrogen atoms from the side chains; nine hydrogen atoms in the case of 1,3,5-trimethyl-

Table 6

Reactions for the decay of anthracene and formation of acenaphthylene, indene and naphthalene.

Rxn	Reaction ^a	A	n	Ea	Ref.
R74	$A_3CH_3 + H = A_3 + CH_3$	6.70E+13	0	6458	[21] ^b
R-74	Reverse	1.20E+12	0	15900	[15] ^c
R75	$A_3 + H = A_3^- + H_2$	2.50E+14	0	15896	[64]
R76	$A_3 + OH = A_3^- + H_2O$	1.70E+12	1.42	1496	[64]
R77	$A_3^- + H = A_3$	1.00E+14	0	0	[64]
R78	$A_2C_2H^* + C_2H_2 = A_3^-$	4.67E+06	1.787	3262	[65]
R79	$A_2^- + C_4H_2 = A_3^-$	3.30E+33	-5.7	25334	[64]
R80	$A_2^- + C_4CH_4 = A_3 + H$	3.30E+33	-5.7	25334	[64]
R81	$INDENYL + C_5H_5 = A_3 + 2H$	5.00E+12	0	8000	[65]
R82	$A_2^- + H = A_2$	1.00E+14	0	0	[66]
R83	$A_2C_2H^* + H = A_2R5$	5.00E+13	0	0	[64]
R84	$INDENE = INDENYL + H$	1.10E+15	0	77096	[64]
R85	$A_3^- + O_2 \Rightarrow CO + HCO + A_2R5$	2.00E+11	0	7352	[64] ^d
R86	$INDENYL + H_2CCCH = A_2R5 + H_2$	9.50E+12	0	0	[64]
R87	$A_2^- + C_2H_2 = A_2A_2R5 + H$	1.90E+31	-5.26	20863	[64]
R88	$A_2C_2H + H = A_2R5 + H$	4.60E+37	-7.03	22949	[64]
R89	$A_2R5^- + H = A_2R5$	1.00E+14	0	0	[64]
R90	$A_2R5 + O = A_2R5^- + OH$	2.00E+13	0	14704	[64]
R91	$A_2R5 + O \Rightarrow A_2^- + HCCO$	2.00E+13	0	14704	[64]
R92	$A_2R5 + H = A_2R5^- + H_2$	2.50E+14	0	15896	[64]
R93	$A_2R5 + OH = A_2^- + CH_2CO$	1.00E+13	0	9935	[64]
R94	$A_2 + O \Rightarrow INDENYL + CO + H$	3.60E+14	0	43899	[64]
R95	$A_2 + O = A_2^- + OH$	2.00E+13	0	14704	[64]
R96	$A_2 + H = A_2^- + H_2$	2.50E+14	0	15896	[64]
R97	$A_2 + OH = A_2^- + H_2O$	2.10E+13	0	45701	[64]
R98	$A_2 + CH_3 = A_2^- + CCH_4$	2.00E+12	0	14962	[64]
R99	$C_6H_5 + H_2CCCH \Rightarrow INDENE$	3.86E+11	0	1361	[64]
R100	$A_2^- + O \Rightarrow INDENYL + CO$	1.00E+14	0	0	[64]
R101	$A_2^- + HO_2 \Rightarrow INDENYL + CO + OH$	1.00E+13	0	0	[64]
R102	$A_3 + C_2H = A_3C_2H + H$	5.00E+13	0	0	[64]
R103	$A_3^- + C_2H_2 = A_3C_2H + H$	1.20E+26	-3.44	30004	[64]
R104	$DIMERS182 + H = C_6H_5CH_2C_6H_5 + CH_3$	6.70E+13	0	6468	[21] ^b
R-104	Reverse	1.20E+12	0	15900	[15] ^c
R105	$C_6H_5CH_2C_6H_5 = FLUORENE + H_2$	1.00E+08	0	32000	[67]

^a $k = AT^n \exp(-Ea/RT)$; rate constant (units: mol, s, cm³, cal).^b Rate constant same as $C_9H_{12} + H = MXYLENE + CH_3$.^c k same as $\Phi C_2H_5 + CH_3 = MCPHC_2H_5 + H$.^d A-factor $\times (1/10)$.**Fig. 5.** Experimental and modeling profiles for the decay of fuel, [symbols]-experiments, [lines]-modeling results.

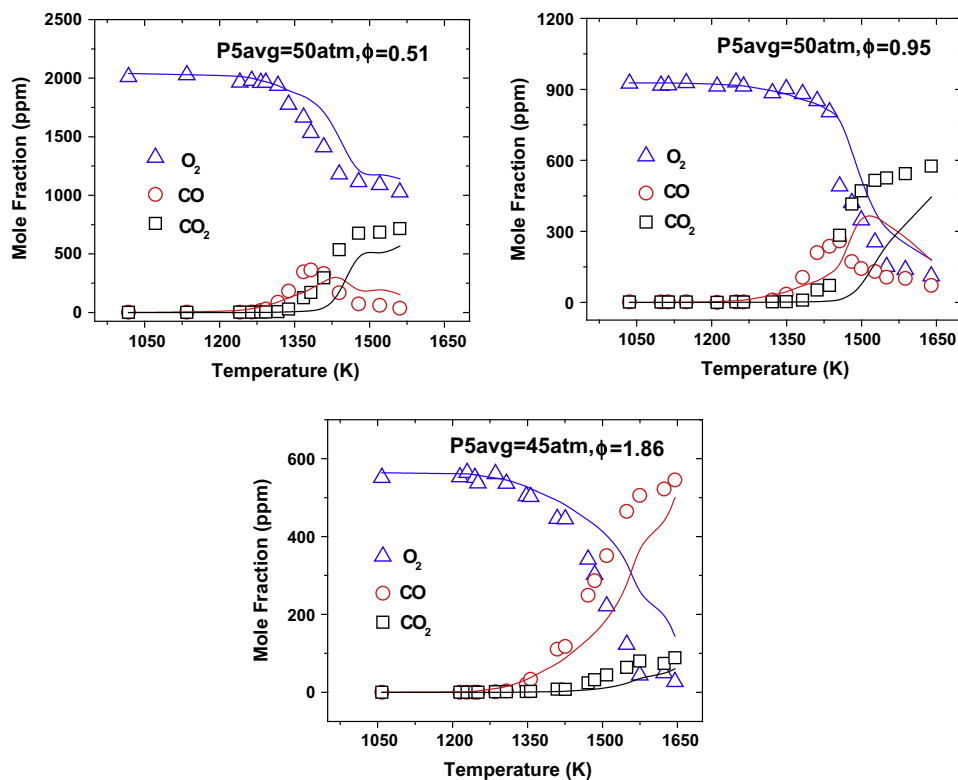


Fig. 6. Experimental and modeling profiles of O_2 , CO and CO_2 . [Symbols]-experiments, [lines]-modeling results.

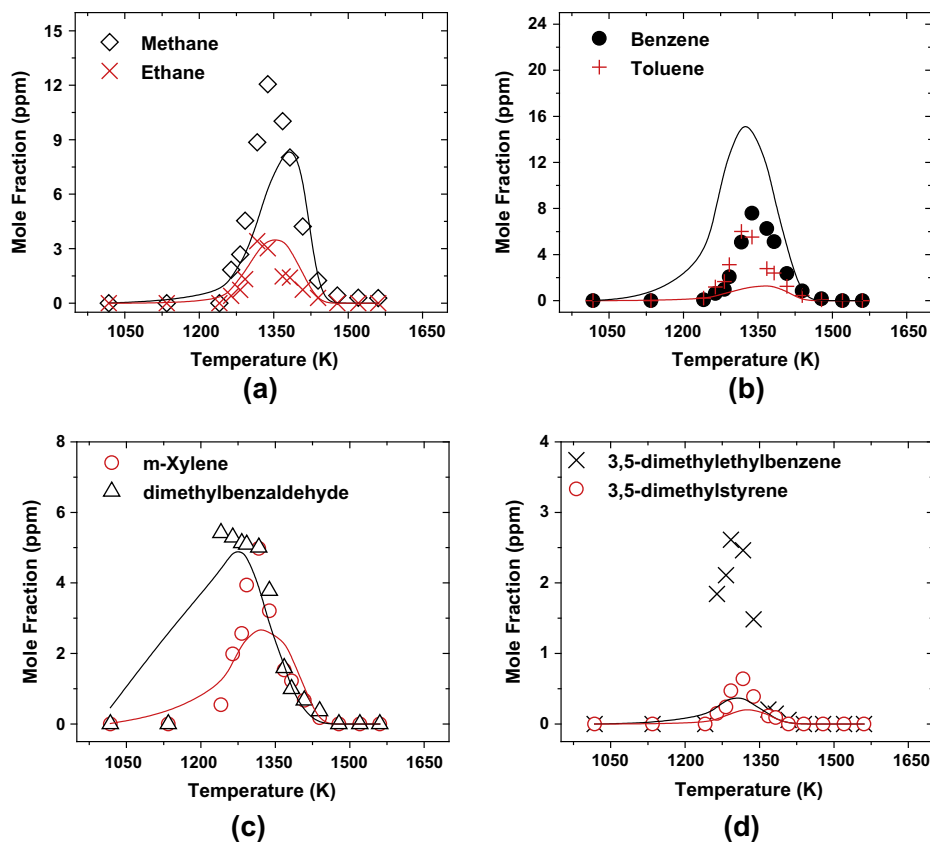


Fig. 7. Experimental and modeling profiles of the major intermediates formed in the oxidation of 1,3,5-trimethylbenzene, [Symbols]- average $P5 = 49.7\text{ atm}$, $\phi = 0.51$, reaction time = 1.26–2.8 ms, [lines]-modeling profiles.

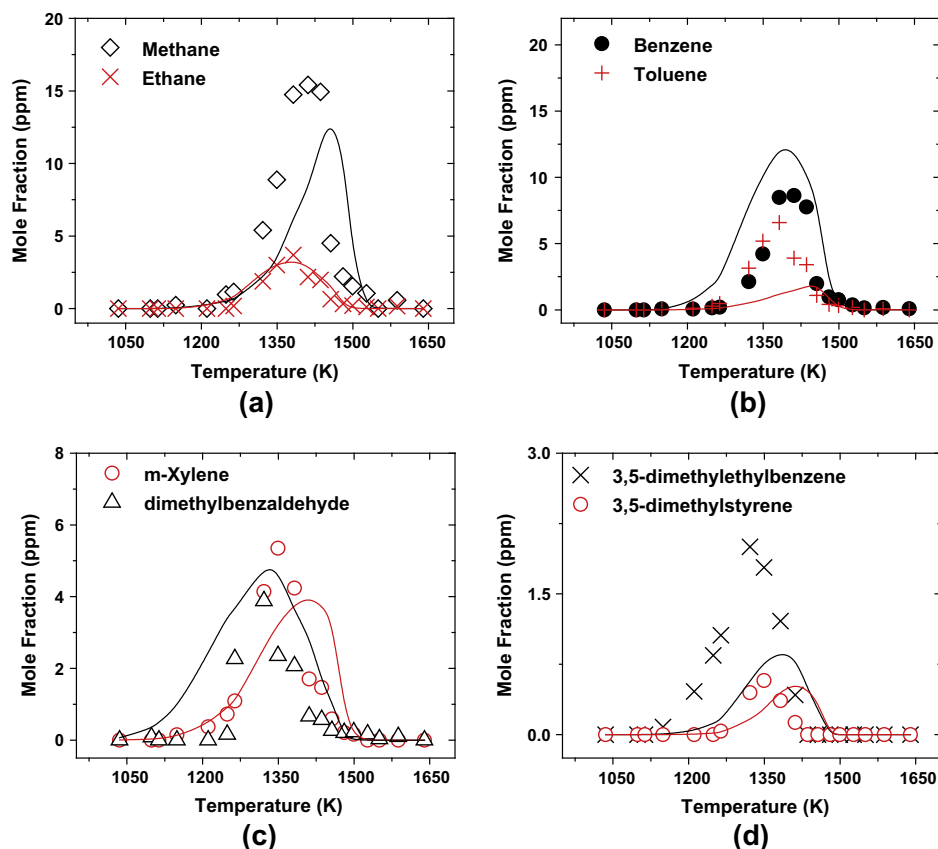


Fig. 8. Experimental and modeling profiles of the major intermediates formed in the oxidation of 1,3,5-trimethylbenzene, [Symbols]-average $P_5 = 50.2$ atm, $\phi = 0.95$, reaction time = 1.13–1.697 ms, [lines]-modeling profiles.

benzene and only three hydrogen atoms in the case of toluene. Toluene has been chosen as the reference species because the oxidation and pyrolytic kinetics of toluene has been extensively studied and direct experimental measurements are available for most of the reactions for the oxidation of toluene [20–41].

The kinetics of the oxidation of 1,3,5-trimethylbenzene are estimated from toluene and such kind of approach has been previously followed in developing chemical kinetic models for oxidation of several aromatic hydrocarbons like *m*-xylene [19,42,43], *n*-propylbenzene [44,45,5] and 1,2,4-trimethylbenzene [46]. The models of *m*-xylene and *n*-propylbenzene oxidation were successfully validated against the speciation data obtained in shock tubes and jet stirred reactor and the ignition delay data obtained in shock tubes. Bikas [46] developed a 1,2,4-trimethylbenzene oxidation model based on toluene and rates of similar reactions have been adjusted to account for higher reactivity of 1,2,4-trimethylbenzene against toluene. A surrogate fuel model was developed Honnet et al. [16] by coupling the 1,2,4-trimethylbenzene mechanism developed by Bikas with *n*-decane mechanism. This combined mechanism of 1,2,4-trimethylbenzene and *n*-decane was used to successfully predict the critical conditions of autoignition of kerosene in non-premixed flows and also volume fraction of soot formed during combustion [16]. Recently the mechanism developed by Honnet et al. [16] which includes the 1,2,4-trimethylbenzene mechanism was used to simulate the extinction strain rates of 1,2,4-trimethylbenzene diffusion flames [5], laminar flame speeds of Jet-A1/air mixtures [47], extinction stretch rate calculations of Jet-A flames [49] and ignition delay times of kerosene/air [48], Jet A mixtures behind reflected shock waves [50].

As mentioned before, the current study is the first detailed experimental and numerical study on the oxidation of 1,3,5-trimethylbenzene at high temperature and pressure conditions and we

were successful in measuring an entire spectrum of products, with carbon numbers ranging from C1 to C16. Most of the previous experimental studies were limited to atmospheric pressure and were conducted for temperatures less than 1000 K. Hence the kinetic data available in literature for the oxidation of 1,3,5-trimethylbenzene is limited and therefore we derived the rate constants for oxidation of 1,3,5-trimethylbenzene from analogous reactions of toluene. Both the forward and reverse rate constants were estimated from analogous reactions of toluene. The reverse rate constants were not calculated from the thermodynamic data provided to chemkin but were explicitly mentioned. The source from which the rate constant has been derived is provided in the column labeled Ref. of the table and the superscripts beside the reference will direct the reader towards the footnotes where any modifications made to the rate constants are listed. Most of the estimated rate constants were not tuned to fit the data better; and any such adjustments made were clearly mentioned in the foot notes below the table. The rate constants for the reactions involved in the oxidation of 1,3,5-trimethylbenzene and the formation of the intermediates are listed in Table 3. The structures of the species in Table 3 could be found in Table 4.

The thermochemistry of the 1,3,5-trimethylbenzene and the intermediate species of the oxidation were estimated by using both group additivity and electronic structure calculations [51]. The enthalpy and entropy of stable intermediates were obtained using the NIST Structures and Properties Group Additivity Model [52]. The enthalpy and entropy of the radical species were determined using density functional theory (B3LYP) calculations with a minimal basis set (6-31G(d)), in view of the computational time and resources needed. The FITDAT utility in Chemkin 3.7[54] was

used to generate the NASA polynomials of the stable intermediates and radicals. The table of thermodynamic properties of the stable and radical intermediate species formed in 1,3,5-trimethylbenzene oxidation is provided in the [Supplementary information](#).

The thermodynamic data is provided to Chemkin so that the reverse rate constants can be calculated. In our case, we provided the reverse rate constants for 1,3,5-trimethylbenzene reaction subset in the model, using the keyword REV. When Chemkin encounters the keyword REV it overrides the thermodynamic data provided in the Chemkin thermo file and uses the provided reverse rate constants for the simulations. Hence the thermodynamic data of 1,3,5-trimethylbenzene and the intermediates of 1,3,5-trimethylbenzene oxidation have not been used in the simulations and only served as dummy variables. The results will be different if the provided thermodynamic data, calculated using B3LYP/6-31G(d) was used to calculate the reverse rate constants. Since the forward rate constants were estimated from toluene, we also estimated the reverse rate constants also from toluene so that the errors due to the thermodynamic data in calculating the reverse rate constants will not be encountered.

The homogeneous batch reactor subroutine in Chemkin10101 [53] was used to simulate our experimental data. As discussed in our previous publication addressing the effect of temperature and pressure on the simulation results [10], for shock tube experiments with less than 15% endwall pressure increase the conventional assumptions of constant pressure and constant temperature leads to reasonable accuracy in predicting the stable species profiles. Hence, the exact pressure, temperature and reaction time were specified for each shock to perform the simulations. All the pressure profiles considered for these simulations had less than 15% endwall pressure increase. The influence of reactions

occurring in the quenching period have insignificant effect on the concentration of the stable species, as discussed in our previous publication [10], hence the quench was not simulated.

4.2. Kinetics for the formation of major two ringed, three ringed and four ringed intermediates from 1,3,5-trimethylbenzene

The major two ringed aromatic hydrocarbons measured in our experiments were 1-(phenylmethyl)-3,5-dimethylbenzene (DIMERS196), 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene ($C_{18}H_{22}$). The major polycyclic aromatic hydrocarbon intermediates measured in our experiments were indene (INDENE), naphthalene (A_2), anthracene (A_3) and methylantracene (A_3CH_3) for all the oxidation experiments. Considerable concentrations of fluorene, cyclopentaphenanthrene, methyleneindene and ethenylantracene were measured for the fuel rich oxidation experiments ($\phi = 2$). The profiles for the formation of the major polycyclic aromatic hydrocarbon intermediates are provided in [Fig. 4](#).

A list of all the other polycyclic aromatic hydrocarbon intermediates which were identified is provided in the experimental data tables of the [Supplementary information](#). The concentrations of these intermediates were considerably small, so we were not successful in obtaining profiles for the formation and consumption of these intermediates. Nevertheless, the identification of these intermediates provides valuable information about the polycyclic aromatic hydrocarbons that are formed from the oxidation of 1,3,5-trimethylbenzene, for our experimental conditions. With the current experimental apparatus, it was not possible to differentiate between various isomers of three ringed aromatic hydrocarbons. Hence, the three ringed aromatic hydrocarbons have been assumed to be anthracene and methylated anthracene species.

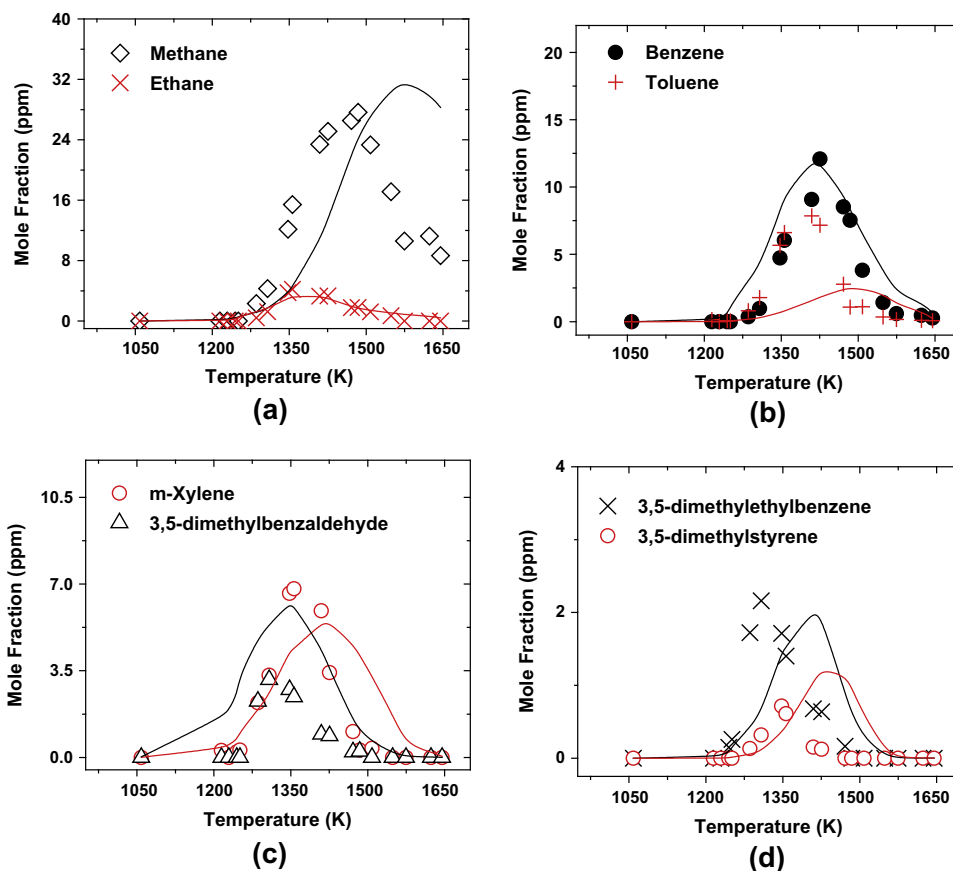


Fig. 9. Experimental and modeling profiles of the major intermediates formed in the oxidation of 1,3,5-trimethylbenzene, [Symbols]- average $P_5 = 45.65$, $\phi = 1.86$, reaction time = 1.11–2.11 ms, [lines]-modeling profiles.

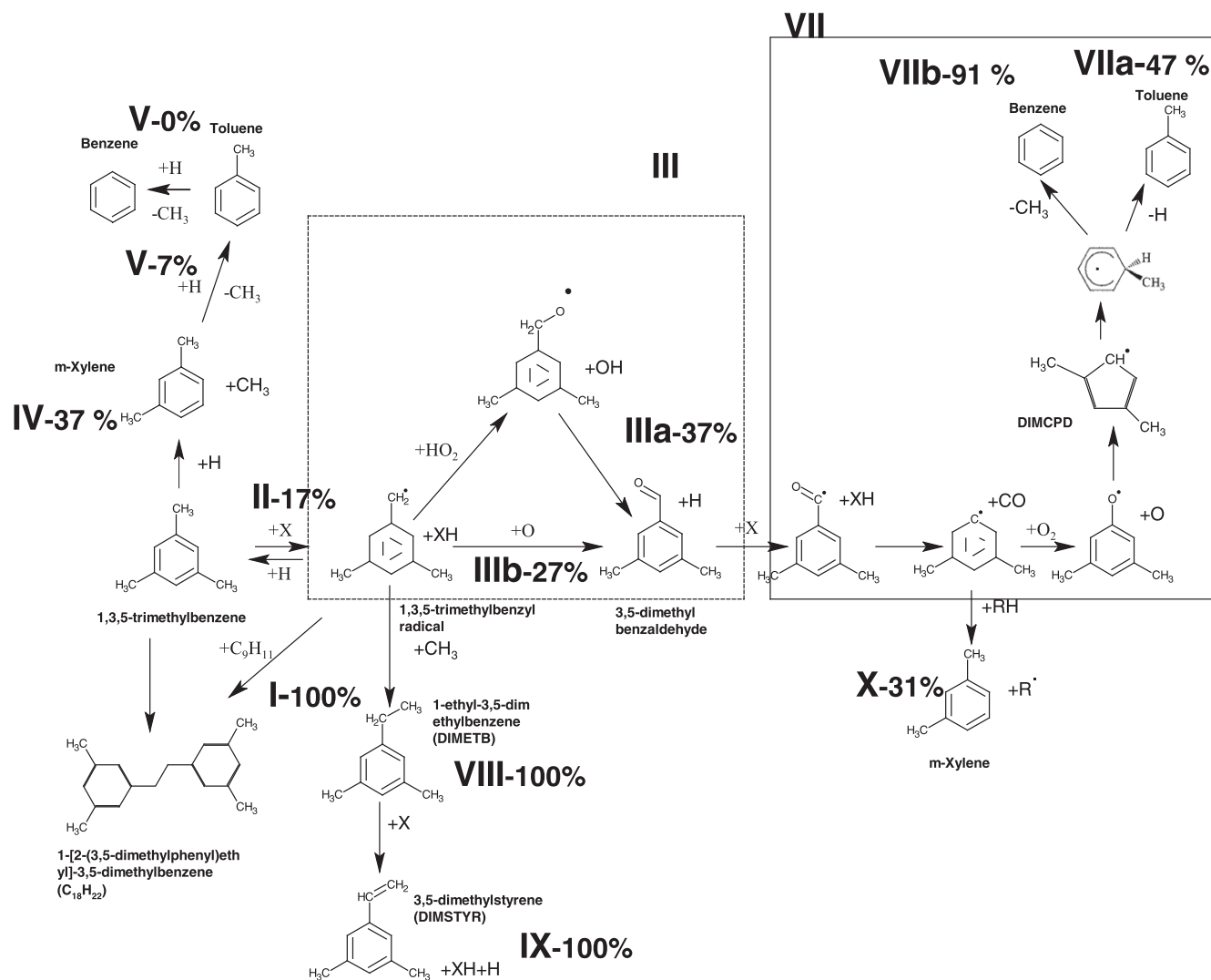


Fig. 10. Reaction path analysis of the fuel and other major intermediates, $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms, $\phi = 0.51$.

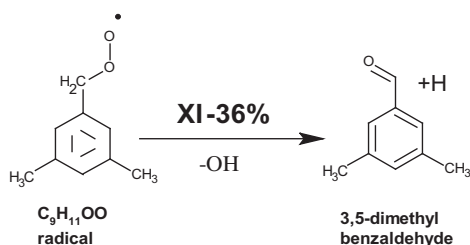


Fig. 11. Reaction path analysis of 3,5-dimethylbenzaldehyde, $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms.

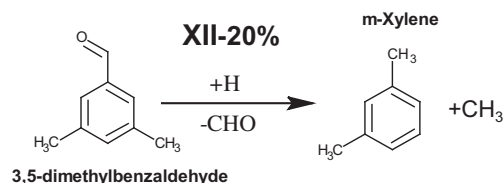


Fig. 12. Reaction path analysis of m-xylene, $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms.

The reaction mechanism and the rate constants for the formation of three ringed aromatic hydrocarbons were included from the publication of Fusetti et al. [15]. In this paper, the authors studied the pathways leading to the formation of methane from methylated monoaromatics, including 1,2,4-trimethylbenzene by conducting pyrolysis experiments in a temperature range of 668–723 K and at a pressure of 100 bar. The authors proposed the mechanism for the formation of various three ringed aromatic hydrocarbons (anthracene) with different degrees of alkylation and from a trimethylbenzene (and not specifically 1,2,4-, 1,2,3- or

1,3,5-trimethylbenzene). Hence these pathways have been incorporated into our kinetic model.

The reactions for the formation of other major polycyclic aromatic hydrocarbons like indene, naphthalene, acenaphthylene and ethenylantracene were included from the atmospheric pressure soot model of Slavinskaya et al. [64], MIT atmospheric pressure soot model [65] Slavinskaya and Frank [66] and reference 67. This reaction subset is provided in Table 6. It was not possible to include reactions for the formation of methyleneindene and cyclopentaphenanthrene in the model due to the limited informa-

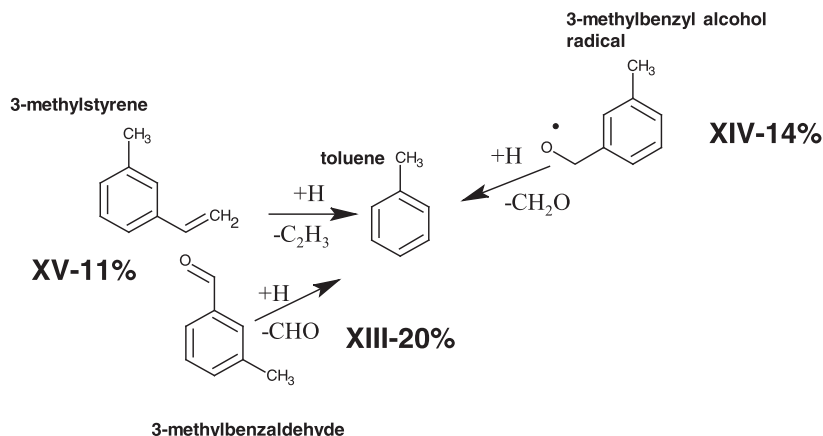


Fig. 13. Reaction path analysis of toluene, $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms.

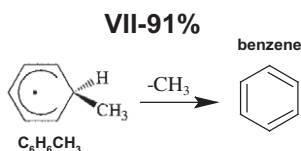


Fig. 14. Reaction path analysis of benzene, $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms.

tion available in the literature for the formation and consumption of these species.

5. Discussion of the Modeling Results

The modeling results of the fuel, oxygen and the major intermediates for the experimental datasets at nominal reflected shock pressure of 50 atm and for three different equivalence ratios ($\phi = 0.51, 0.95$ and 1.86) are shown in Fig. 5 through Fig. 9. The experimental and modeling profiles for the fuel lean and fuel rich oxidation datasets at a nominal pressure of 20 atm could be found in the Supplementary information. The experimental data is shown by open and closed symbols and the modeling results are shown by straight or dashed or dotted lines.

5.1. Interpretation of modeling of the fuel decay and formation of methane, ethane and major single ringed aromatic hydrocarbons

The experimental and modeling profiles of the fuel, O₂, CO and CO₂, for three different equivalence ratios ($\phi = 0.51, 0.95$ and 1.86) and for a nominal reflected shock pressure of 50 atm, are presented in Fig. 5 and Fig. 6. The modeling profiles show good agreement with the decay of the fuel. The experimental and modeling profiles for the decay of the oxygen and the formation of CO and CO₂ are shown in Fig. 6. The modeling predictions show good agreement

for the decay of the oxygen and the formation of CO and CO₂ for all the three experimental datasets.

The results of the simulation are shown in Figs. 7–9 against the experimental data, for other major intermediates such as methane, ethane, benzene, toluene, m-xylene, dimethylbenzaldehyde (DIM-PHHCO), 3,5-dimethylethylbenzene (DIMETB) and 1-ethenyl-3,5-dimethyl benzene (DIMSTYR) for the datasets at a nominal pressure of 50 atm and for three different equivalence ratios ($\phi \approx 0.5, 1$ and 2). The modeling results show peak concentrations of these intermediates at the correct experimental temperatures which indicate that the hypothesized pathways and the estimated kinetics for the consumption of the fuel and formation of the intermediates are accurate.

The proposed 1,3,5-trimethylbenzene oxidation mechanism, which is analogous to m-xylene oxidation mechanism [18] is shown in Fig. 10. Reaction path analysis of the fuel and the major intermediates was performed for a temperature of 1381 K, reaction time of 1.58 ms, and pressure of 46 atm and at fuel lean conditions to validate our hypothesized mechanism and these results are also shown in Fig. 10. The direction of the arrow indicates the contribution of that specific pathway to the formation of the species shown below the arrow. For example, the formation of 1-ethyl-3,5-dimethylbenzene is shown by pathway VIII in Fig. 10. It is formed from the recombination reaction of 1,3,5-trimethylbenzyl and methyl radicals. The number 100% beside VIII indicates that this pathway is the only contributor to the formation of 1-ethyl-3,5-dimethylbenzene. A few interesting aspects could be observed from Fig. 10 and these points (1) and (2) are discussed in greater detail in the following sections (Sections 5.1.1 and 5.1.2).

- (1) Less than 70% of toluene (V and VIIa), m-xylene (IV and X) and 3,5-dimethylbenzaldehyde (IIIa and IIIb) is formed from the pathways shown in the figure, which implies that there are other pathways (not shown in Fig. 9) which contribute to the formation of these species.

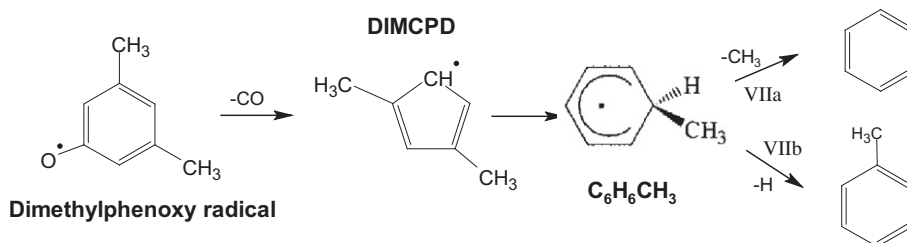


Fig. 15. Reaction mechanism of dimethylphenoxy radical.

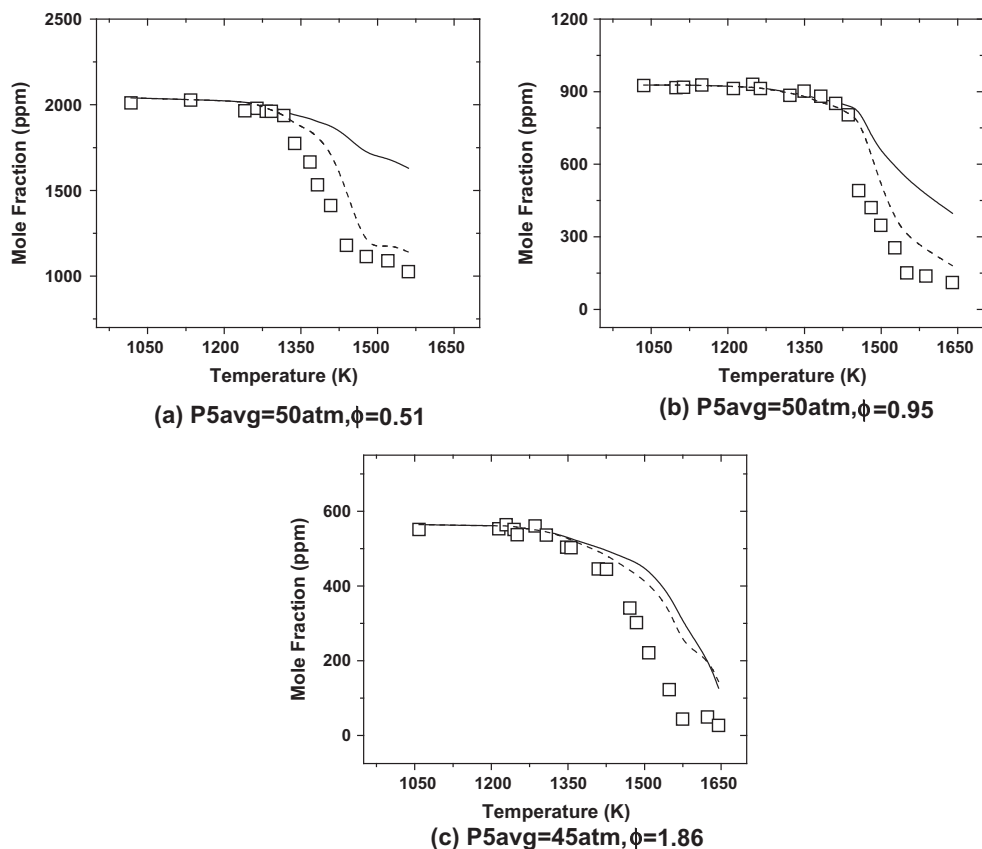


Fig. 16. Experimental and modeling profiles of oxygen, [---]—modeling results including the chemistry of cyclic C6 species, [—]—modeling results without including the chemistry of cyclic C6 species.

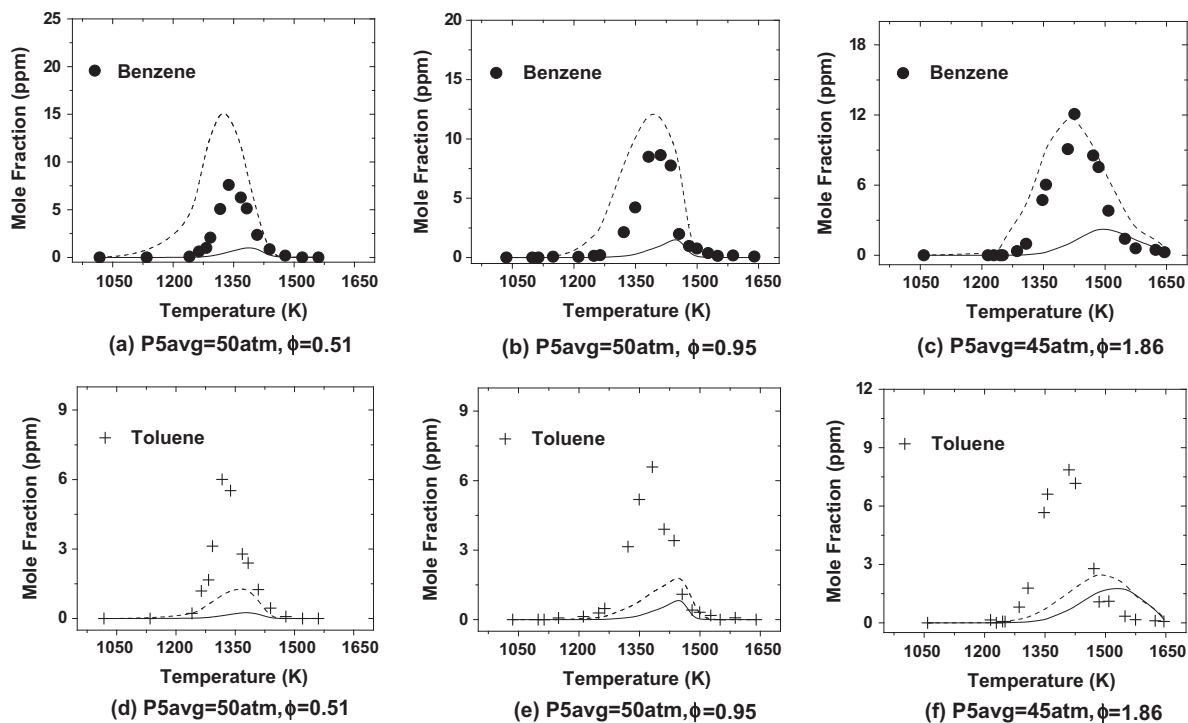


Fig. 17. Experimental and modeling profiles of benzene and toluene, [---]—modeling results including the chemistry of cyclic C6 species, [—]—modeling results without including the chemistry of cyclic C6 species.

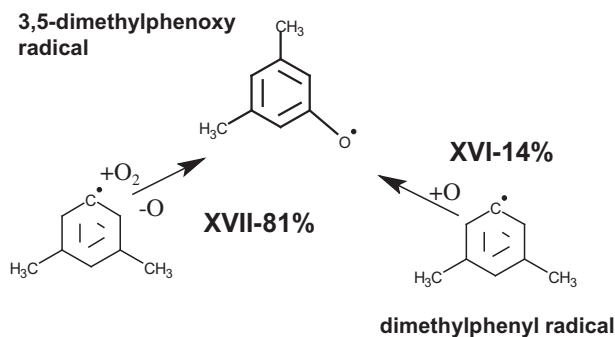


Fig. 18. Formation of dimethylphenoxy radical in 1,3,5-trimethylbenzene oxidation, reaction path analysis at $P_5 = 46$ atm, $T_5 = 1381$ K, reaction time = 1.58 ms.

- (2) Pathway V shows that 0% of benzene is formed from toluene. About 91% of benzene is produced from dimethylphenoxy radical (VIIb).

5.1.1. Importance of $C_9H_{11}+O_2$, $C_9H_{11}+HO_2$ and $C_9H_{11}+O$ reactions in the formation of intermediates

Additional pathways have been included in the model apart from those proposed in Fig. 10 so as to build a comprehensive kinetic model. These additional pathways include reactions of C_9H_{11} radical with HO_2 (reactions R40 to R42 in Table 3), O_2 (reactions R27–R29 in Table 3) and O species (reaction R37 in Table 3) and reactions for the formation of smaller carbon number alkylated single-ring aromatic hydrocarbons like m-xylene and toluene from di-alkylated and tri-alkylated single-ring aromatic intermedi-

ates (reactions R23–R26 and R47–R52 in Table 3). The species, m-xylene, 3,5-dimethylbenzaldehyde and toluene are partially formed from the reactions shown in Fig. 10 and are partly formed from these additional pathways.

About 36% of 3,5-dimethylbenzaldehyde is formed from $C_9H_{11}OO$ radical (XI in Fig. 11). The $C_9H_{11}OO$ radical is formed from the reaction of C_9H_{11} and O_2 .

About 20% of m-xylene is formed from 3,5-dimethylbenzaldehyde by displacement of the formyl side chain by the hydrogen atom (XII in Fig. 11). About 45% of toluene is formed from 3-methylbenzaldehyde, 3-methylbenzylalcohol radical and 3-methylstyrene by displacement of the non-methyl side chains by a hydrogen atom (see pathways XIII, XIV and XV in Fig. 13).

From the pathways shown in Figs. 11–13, we can deduce that the oxidation of C_9H_{11} radical by O_2 , HO_2 and O radicals is important and influences the formation of the stable intermediates such as 3,5-dimethylbenzaldehyde and m-xylene. Toluene and m-xylene are formed from dialkylated and trialkylated single ring aromatics via displacement of the non-methyl side chains by a hydrogen atom.

5.1.2. Importance of dimethylphenoxy radical chemistry in the formation of benzene

Unlike m-xylene and toluene, benzene is not formed from toluene, styrene, benzaldehyde or benzylalcohol through displacement of the side chains by a hydrogen atom. About 91% of benzene is formed from cyclic species, $C_6H_6CH_3$. Hence, this cyclic species is the primary contributor to the formation of benzene.

We have previously noticed the importance of the cyclic compound $C_6H_6CH_3$ in the formation of benzene and toluene in our m-xylene oxidation experiments [19]. In m-xylene oxidation, the

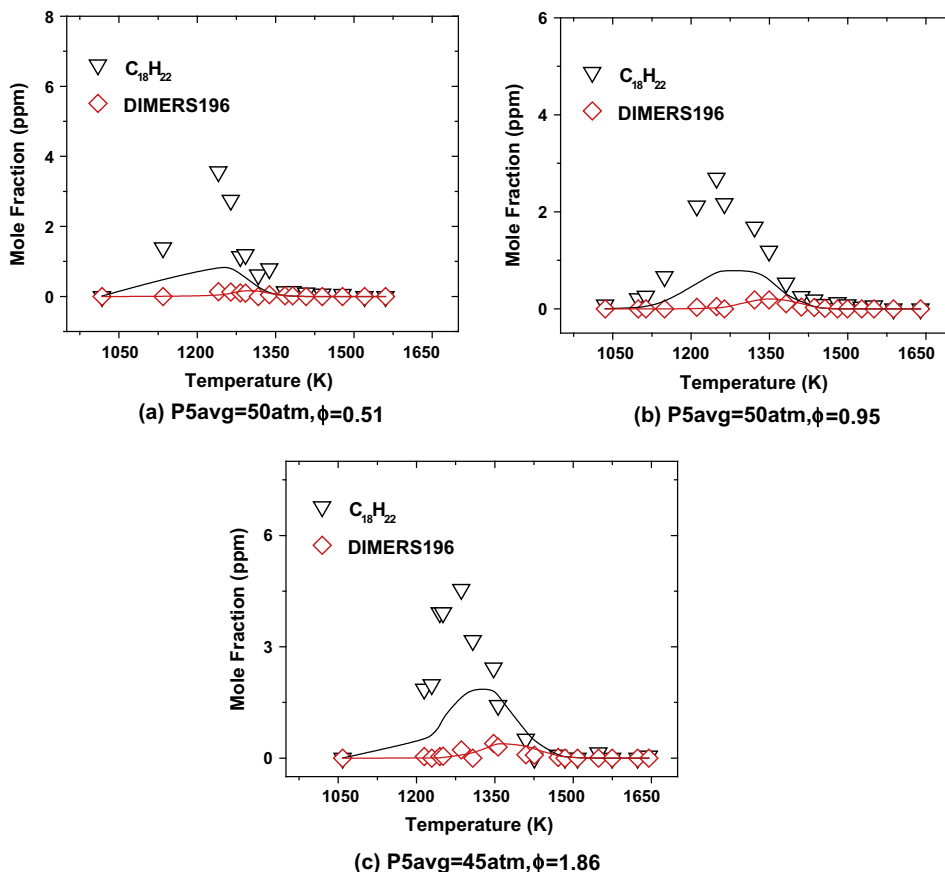


Fig. 19. Experimental and modeling profiles of 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene ($C_{18}H_{22}$) and 1-(phenylmethyl)-3,5-dimethylbenzene (DIMERS196).

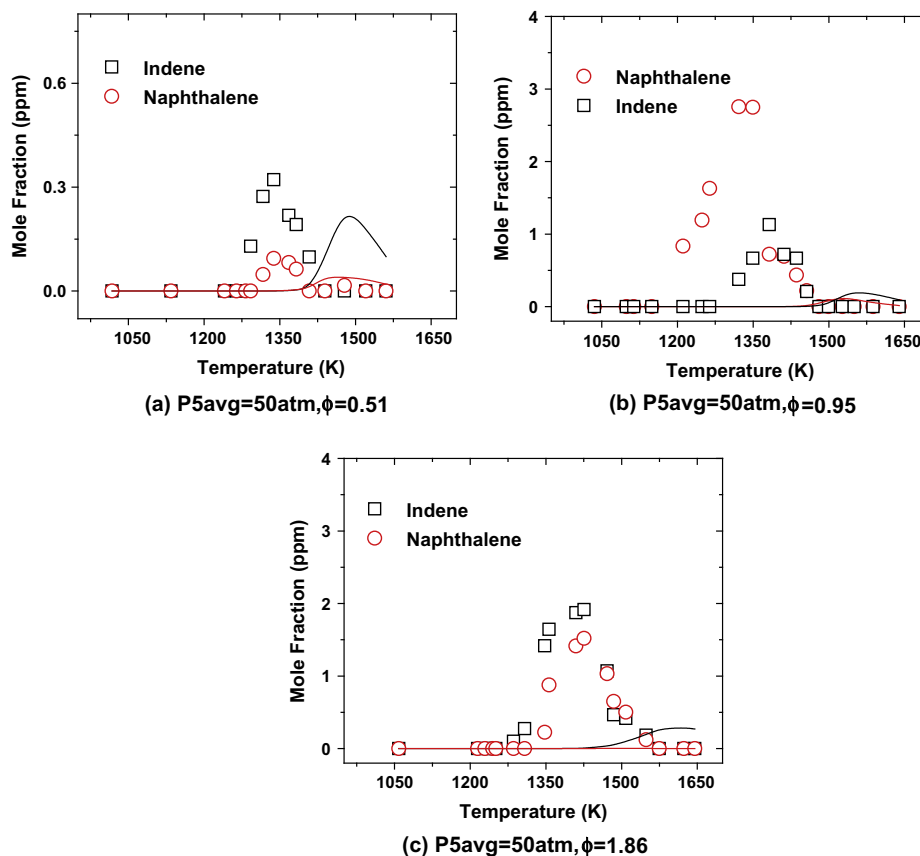


Fig. 20. Experimental and modeling profiles of indene and naphthalene.

$C_6H_6CH_3$ radical is produced from dimethylphenoxy radicals and the decay of $C_6H_6CH_3$ species leads to the formation of benzene and toluene (mechanism shown in Fig. 15). In order to emphasize the importance of this pathway even in the oxidation of 1,3,5-trimethylbenzene, a simple modeling experiment was conducted by removing the reactions for the formation of benzene and toluene from $C_6H_6CH_3$ (VIIa and VIIb in Fig. 15). The results of the simulation are shown in Fig. 16 and Fig. 17. The modeling results for the formation of benzene, toluene and oxygen in 1,3,5-trimethylbenzene oxidation experiments by including these reactions are shown by dashed lines and the modeling results after removal of these reactions are shown by straight lines. From Fig. 16 we can observe that removal of these reactions significantly under predicts the consumption of oxygen for the fuel lean and stoichiometric experimental datasets and the formation of benzene for all the experimental datasets (see Fig. 17a–c). We can see significantly lower concentrations of benzene being formed, when these reactions are not included. Similarly, the concentrations of toluene are under predicted when these pathways are not included (see Fig. 17d–f). Hence, this modeling experiment reveals the importance of including the dimethylphenoxy radical chemistry in the modeling of not only m-xylene oxidation but also in the oxidation of 1,3,5-trimethylbenzene. In the case of 1,3,5-trimethylbenzene the dimethylphenoxy radicals are formed from the oxidation of dimethylphenyl radical (XVI and XVII in Fig. 18).

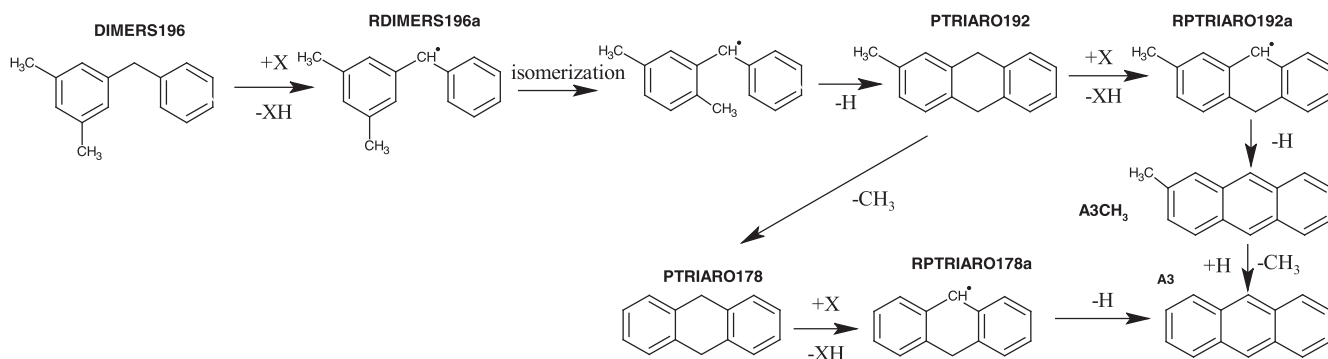
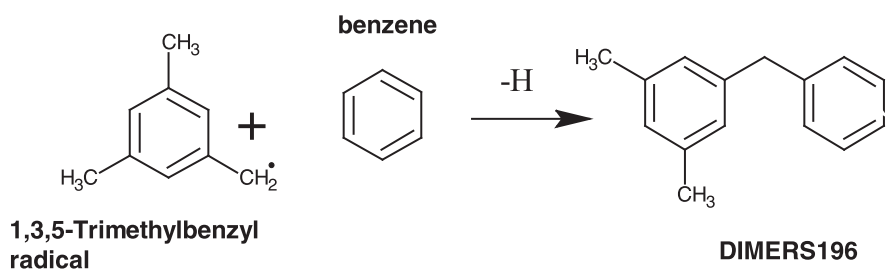
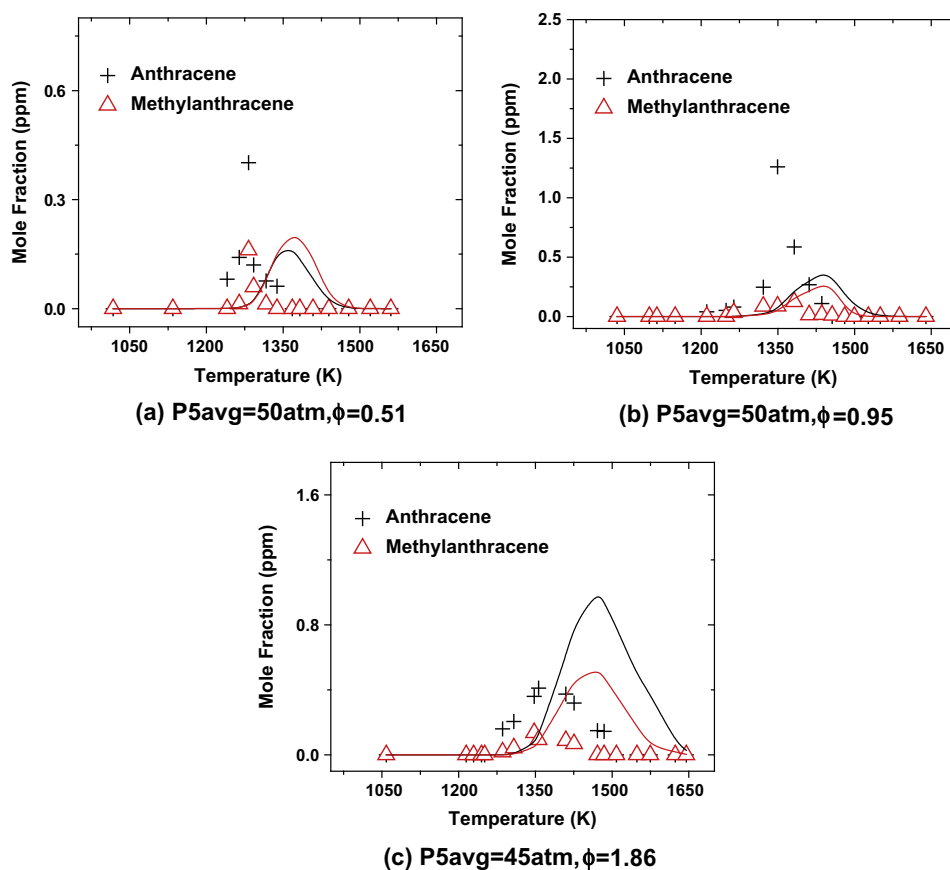
From Fig. 17a–c, it can be observed that the peak modeling concentrations of benzene vary from about 15 ppm for the fuel lean dataset to 12 ppm for the fuel rich dataset. The peak concentrations of benzene in the experiments vary from 7 ppm to 12 ppm. Reaction path analysis for the formation of benzene at the peak experimental concentrations showed that benzene is primarily formed from $C_6H_6CH_3$ species (see Fig. 14), for all the experimental

datasets ($\phi = 0.51, 0.95$ and 1.86). The species, $C_6H_6CH_3$ is formed through a sequence of steps following the oxidation of dimethylphenyl radical to dimethylphenoxy radical (see Fig. 15 and Fig. 18). Hence at lower equivalence ratios higher amounts of benzene are formed in the modeling concentrations when compared to fuel rich conditions due to oxidation of the dimethylphenyl radical to $C_6H_6CH_3$ species. The oxidation of dimethylphenyl radical to dimethylphenoxy radical is the only step included in the model for the consumption of dimethylphenyl radical. Including additional steps for the consumption of dimethylphenyl radical, which results in products other than benzene, could produce lower amounts of benzene in the modeling results at fuel lean and stoichiometric conditions. However the nature of these additional steps are not known at present and hence were not included in the model. As the equivalence ratio is increased the contribution of the step shown in Fig. 15 is decreased which results in lower amounts of benzene to form in the simulation results.

The model shows very low concentrations of toluene being formed, in spite of having reactions for displacement of the non-alkyl side chain by a hydrogen atom, from di-alkylated aromatics like m-xylene, 1-ethyl-3-methylbenzene and 3-methylbenzaldehyde. This implies that an alternative pathway exists for the formation of toluene from 1,3,5-trimethylbenzene.

If we compare the modeling profiles for the formation of toluene and benzene (shown by straight lines), in Fig. 17, we can observe the similarity in the peak shapes for the simulation results of these two intermediates, in the absence of the chemistry of the cyclic $C_6H_6CH_3$ species. This observation confirms our hypothesis that an alternative pathway exists for the formation of toluene, which has not been included in the model.

Several publications on atmospheric kinetics have observed the formation of alkylated phenols in the oxidation of aromatics [68],



such as toluene [69,70] and trimethylbenzene [72,73]. Since our experiments are conducted at much higher pressures the mechanism of decay of the trimethylphenol is probably not similar to what is postulated in these publications. However, similar to the decay of dimethylphenoxy radical which results in the formation of benzene and toluene, the trimethylphenoxy radical could result in the formation of minor amounts of m-xylene and major amounts of toluene. The trimethylphenoxy radical could be formed from the oxidation of trimethylphenyl radical. The pathways and kinetics for the formation of m-xylene and toluene from trimethylphenoxy radical is at present not clear to us. Hence, these pathways have not been included into the model. We hypothesize that inclusion of this pathway will result in increased amounts of toluene being formed in the modeling concentrations.

The reaction path analysis of the major intermediates validates our hypothesized mechanism for the oxidation of 1,3,5-trimethylbenzene. Hence, the oxidation of 1,3,5-trimethylbenzene indeed takes place by sequential oxidation and removal of methyl side chains. However, the decreased concentrations of toluene observed in the modeling results indicate the possibility of having alternative pathways for the formation of this species which do not fit into our generic description of 'oxidation of 1,3,5-trimethylbenzene by sequential oxidation and removal of the methyl side chains'.

5.2. Interpretation of the formation of two ringed and three ringed aromatic hydrocarbons

The experimental and modeling profiles of indene, naphthalene, anthracene, 1-(phenylmethyl)-3,5-dimethylbenzene (DIMERS196), 1-[2-(3,5-dimethylphenyl)ethyl]-3,5-dimethylbenzene ($C_{18}H_{22}$) and methyl anthracene are shown in Figs. 19–21, for three different equivalence ratios ($\phi = 0.51, 0.95$ and 1.86). From Fig. 19a–c, it can be observed that the model showed the formation of DIMERS196 and $C_{18}H_{22}$ at the right experimental temperatures. The model shows good predictions for the formation of DIMERS196, for all the experimental datasets. The modeling concentrations of $C_{18}H_{22}$ are slightly under predicted for almost all the experimental datasets.

Displaced profiles for the formation of indene and naphthalene could be observed in the fuel lean dataset (see Fig. 20a). However, for the stoichiometric and fuel rich dataset significantly low concentrations of indene and naphthalene could be observed (see Fig. 20b and c). From Fig. 21a–c, it can be observed that the model was successful in capturing the formation of anthracene and methyl anthracene at lower temperatures and increased amounts of anthracene and methylanthracene are observed at higher temperatures for all the experimental datasets. The modeling results showed negligible concentrations of fluorene and ethenylanthracene being formed for the fuel rich dataset (the profiles are not shown).

Reaction path analysis of $C_{18}H_{22}$ revealed that this species is formed from the recombination of 1,3,5-trimethylbenzyl radicals (see pathway I in Fig. 10). The species, DIMERS196 (100%) is formed from recombination of 1,3,5-trimethylbenzyl radical with benzene followed by removal of hydrogen atom (Fig. 22).

Anthracene and methylanthracene are confirmed to be formed from DIMERS196 through the sequence of steps shown in Fig. 23, which are similar to the mechanism proposed by Fusetti et al. [15]. The authors studied the thermal cracking of methylated monoaromatics which included 1,2,4-trimethylbenzene and proposed a generic pathway for the formation of methylanthracene from trimethylbenzenes. This generic pathway has been modified by including the isomerization step so that methylanthracene could be formed from DIMERS196 (reaction path analysis was not performed for the formation of anthracene and methylanthracene, since only this reaction sequence is present in the model for

the formation of these species). Displacement of the methyl side chain from anthracene by hydrogen atom produces anthracene.

The lumping of all the trimethylbenzenes has been adopted only for the reaction mechanism shown in Fig. 23. These reactions are shown in Table 5 and are 30 in number. The 1,3,5-trimethylbenzene oxidation model subset which was appended to the UIC m-Xylene Oxidation Model 2 consists of 201 reactions (including the reverse reactions). Hence the lumping approach has been followed only for 30 reactions of 201 reactions of the 1,3,5-trimethylbenzene oxidation model subset. This approach has been followed because no other study in literature, other than that of Fusetti et al. [15] provided rate constants for the formation of three ringed species from a two-ringed intermediate (DIMERS196) and including these reactions steps was important in simulating the formation of anthracene and methylanthracene which were quantified in our experiments.

Reaction path analysis was not performed for indene and naphthalene because of the very low modeling concentrations and displaced profiles for the formation of these intermediates, as of result of which the analysis might not give accurate information for the formation of these species. But the absence of formation of indene and naphthalene in the modeling profiles could be due to the lower concentration of toluene being formed in the simulation results. The importance of the presence of intermediates of toluene oxidation, such as benzyl radical, cyclopentadienyl radical and fulvenallene in the formation of indene and naphthalene has been mentioned previously in many publications [31,74–79]. Improved predictions for the formation of toluene can also result in improved modeling profiles for indene and naphthalene.

6. Concluding remarks

The oxidation of 1,3,5-trimethylbenzene has been studied at nominal reflected shock pressures of 25 and 50 atm, for a temperature range of 1018–1640 K at fuel lean, stoichiometric and fuel rich conditions. Species profiles of small hydrocarbons, mono-aromatic and multi-ringed aromatic species were obtained as a function of temperature. A 1,3,5-trimethylbenzene oxidation model was developed to predict the decay of the fuel and the formation of aliphatic and single ringed aromatic hydrocarbons, as measured from our experiments. The model shows satisfactory predictions for the formation of most of the major intermediates. Through reaction path analysis it was identified that the pathways involving oxidation of the 1,3,5-trimethylbenzyl and dimethylphenyl radical play an important role in the formation of the major intermediates such as benzene, toluene and 3,5-dimethylbenzaldehyde. Polycyclic aromatic hydrocarbon species such as anthracene and methylanthracene were formed through a sequence of steps, following the reaction of 1,3,5-trimethylbenzyl radical with benzene. The oxidation of 1,3,5-trimethylbenzene seems to have additional complications beyond our simple hypothesis of oxidation by simultaneous oxidation and abstraction of the methyl side chains.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2012.06.014>.

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